

Board Manual

for Development Board 2.0.0

VRender™ 3D

User Manual

MDVR3511FDEB2UM-V1

Ver 1.0 / February 6, 2006

VR3511F

Board Manual for Development Board based on VR3511F Processor

Available Board - Development Board 2.0.0

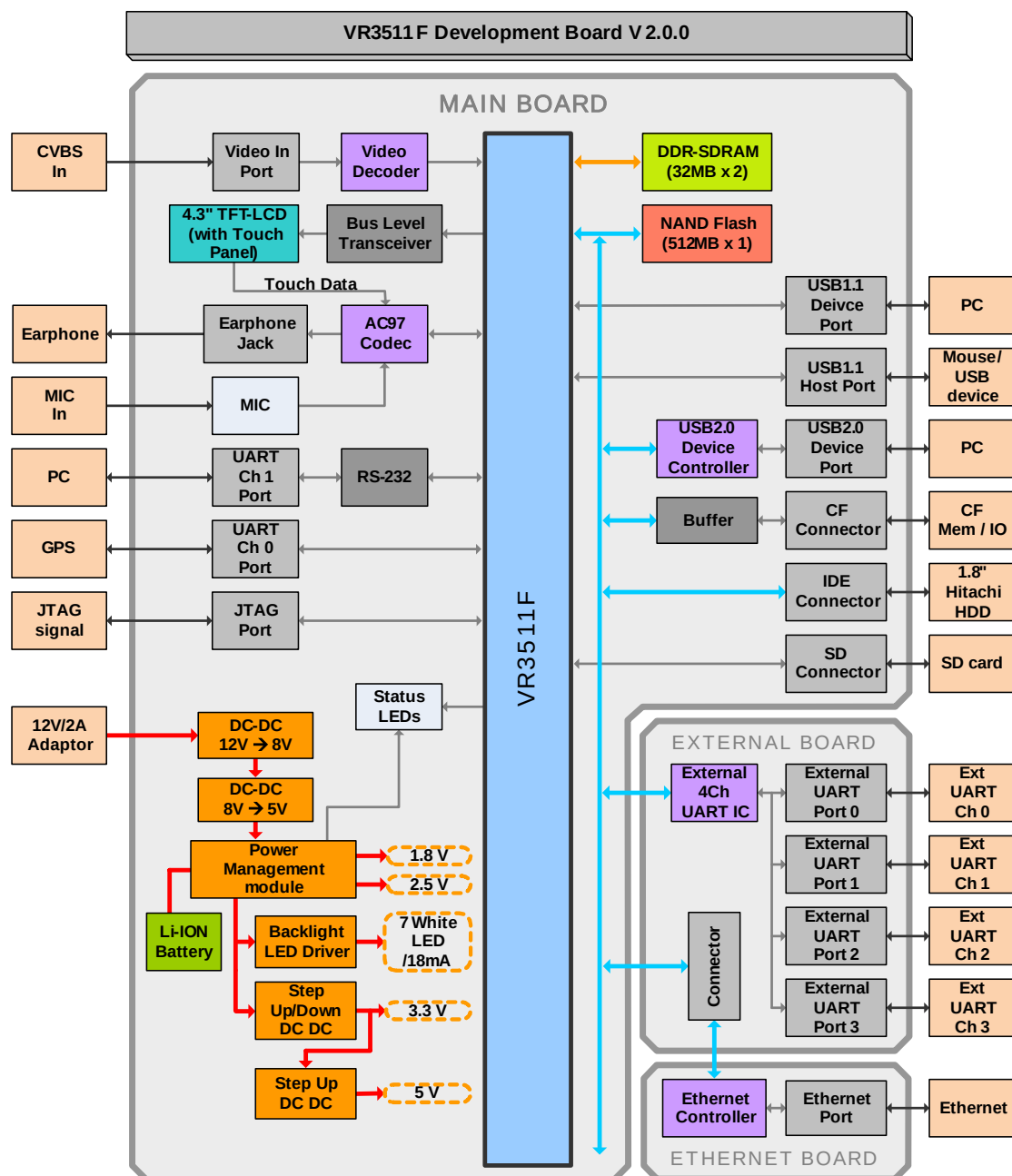
Table of Contents

1.	Introduction.....	3
1.1.	System Block Diagram.....	3
2.	H/W Specification.....	4
3.	Function Description	6
3.1.	System Configuration Pins	6
3.2.	Power Block	7
3.2.1.	Power Management Unit (LTC3455)	8
3.2.2.	Operation of Momentary-On/Off Switch.....	8
3.2.3.	LCD Back light Power control	9
3.2.4.	3.3V Power generation	9
3.3.	JTAG Interface.....	9
3.4.	Reset.....	10
3.5.	Clock	11
3.6.	DDR-SDRAM Memory Interface.....	12
3.7.	NAND Flash Interface	13
3.8.	LCD Interface.....	14
3.9.	Video-In Interface.....	15
3.10.	AC97 Codec Interface.....	16
3.11.	USB2.0 Controller Interface.....	17
3.12.	USB1.1 Host Interface	17
3.13.	USB1.1 Device Interface	18
3.14.	UART Interface	19
3.15.	IDE Interface	19
3.16.	CF Interface	20
3.17.	SD/MMC Interface	22
3.18.	Ethernet Controller Interface.....	23
3.19.	External UART IC Interface	24
3.20.	Key Function	25
4.	Components Description	26
4.1.	Components Arrangement.....	26
4.1.1.	Top View	26
4.1.2.	Bottom View.....	27
4.1.3.	External Board	27
4.1.4.	Ethernet Board.....	28
4.1.5.	Board to board connection.....	28
4.2.	Components Description	29
4.2.1.	Main Board.....	29
4.2.2.	External Board	35
4.2.3.	Ethernet Board.....	37

1. Introduction

VR3511F Development Board 2.0.0 has similar features with car navigation kit and provides easy development condition to the customer. <Fig 1> shows the total block diagram of VR3511F Development Board 2.0.0.

1.1. System Block Diagram



<Fig 1. Block diagram of VR3511F Development Board 2.0.0>

2. H/W Specification

- **Core Processor**

- Part name : MagicEyes VR3511F

- **LCD Display Component**

- 4.3 inch TFT-LCD module
- Part name : LTG430WQ-F02
- Resolution : WQVGA(480x272)
- With Touch Panel

- **Main memory**

- DDR-SDRAM(256Mbit) x 2
- Total memory capacity : 512Mbit
- Part name : K4H561638F-GCB3
- Package type : FBGA

- **Storage memory**

- NAND Flash(4Gbit) x 1
- Block type : Large Block
- Part name : K9K4G08U0M
- Package type : 48pin TSOP1

- **Storage**

1. HDD

- 1.8 inch HDD Interface
- Only support Hitachi 1.8inch Travelstar C4K60 series
- Supported models : HTC426060G8CE00 / HTC426040G8CE00
HTC426030G5CE00 / HTC426020G5CE00

2. SD/MMC

3. Compact Flash

- **USB**

- Internal USB1.1 Host x 1 Channel
- Internal USB1.1 Device x 1 Channel
- External USB2.0 Device(using NET2272) x 1 Channel

- **UART**

1. Default

- Use internal UART of VR3511F
- UART Channel 0 : Tx/Rx UART signal for GPS
- UART Channel 1 : Tx/Rx UART signal for debugging

2. External (Daughter board)

- Use external UART controller IC(ST16C554) in External board
- Ext UART Channel 0 : Full feature UART signal support
- Ext UART Channel 1 : Tx/Rx UART signal support
- Ext UART Channel 2 : Tx/Rx UART signal support
- Ext UART Channel 3 : Tx/Rx UART signal support

- **Video In**

- Support CVBS, Y/C video input format
- Part name : SAA7113H

- **Audio In/Out**

- Stereo Audio Out x 1
- Stereo Audio In x 1
- MIC In x 1
- Part name : WM9712EFL AC97 codec

- **Ethernet (Sub board)**

- Ethernet 10 base x 1
- Part name : CS8900 Ethernet controller IC

- **JTAG Port**

- **Power**

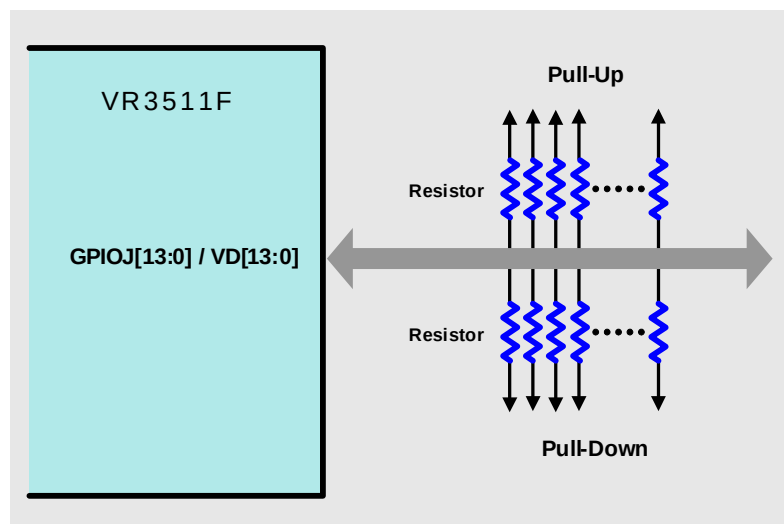
- Adaptor : 12V/2A
- 2200mA Li-ION Battery

3. Function Description

Chapter 3 describes every hardware-related features on VR3511F Development Board 2.0.0.

3.1. System Configuration Pins

VR3511F has System Configuration Pins that configure the system initial value, such as NAND boot, NOR boot, SDRAM bus width, SDRAM capacity, and so on. If a reset signal is asserted to VR3511F, VR3511F will read the value of Configuration Pins and it will be initialized according to the setting value of Configuration Pins. Therefore they must be set to pulled-up or pulled-down correctly. <Fig 2> shows the concept of System Configuration Pin setting. These pins only work as system configuration function pins when reset occurring and after reset, work as GPIO function pins or ALTERNATE(hereafter ALT) function pins according to the setting value of GPIO Alternate Function Registers.



<Fig 2. Concept of System Configuration Pin setting>

Below <Table 1> shows the System Configuration Pin settings of VR3511F Development Board 2.0.0.

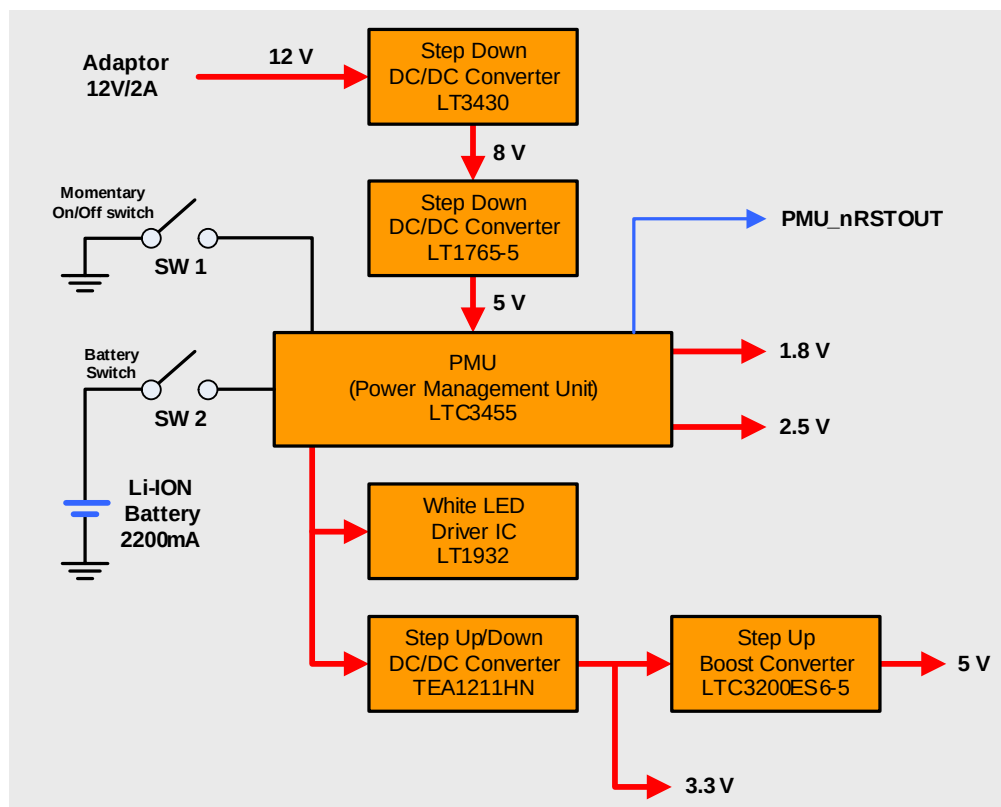
Signal Name	Related Configuration function	Setting Value	Description
VOUT_D0	CfgShadow	1	Shadow mode enable
VOUT_D1	CfgBootType0	0	b'10 : NAND boot
VOUT_D2	CfgBootType1	1	
VOUT_D3	CfgSDRLAT	0	CAS Latency 2 cycle
VOUT_D4	CfgNFBUSWidth	0	8 bit bus width NAND Flash
VOUT_D5	CfgNFType	1	4 address type NAND Flash
VOUT_D6	CfgSR0BUSWidth	1	16 bit bus width SRAM bus

VOUT_D7	CfgSDRTYPE	1	Default setting value. Do not change this value to 0
VOUT_D8	CfgSDRBUSBW0	1	32 bit bus width DDR-SDRAM bank (Z-bank)
VOUT_D9	CfgSDRBUSBW1	1	
VOUT_D10	CfgSDRBW0	0	16 bit bus width DDR-SDRAM IC
VOUT_D11	CfgSDRBW1	1	
VOUT_D12	CfgSDRCAP0	0	256Mbit capacity DDR-SDRAM IC
VOUT_D13	CfgSDRCAP1	1	

<Table 1. System Configuration Setting of VR3511F Development Board 2.0.0>

<Caution 1> The meaning of bank

CfgSDRBUSBW[1:0] means total bus width of DDR-SDRAM bank. This 'bank' means DDR-SDRAM bank of VR3511F(different from the meaning of bank that is mentioned at DDR-SDRAM specification). For example, Samsung K4H561638F is a 256Mbit DDR-SDRAM IC and its structure is 4M x 16bit x 4bank. This 'bank' means an inner bank of DDR-SDRAM IC. Do not confuse 'bank' of VR3511F with 'bank' of DDR-SDRAM IC.

3.2. Power Block

<Fig 3. Block diagram of Power Block>

<Fig 3> shows the power block diagram of VR3511F Development Board 2.0.0. For making

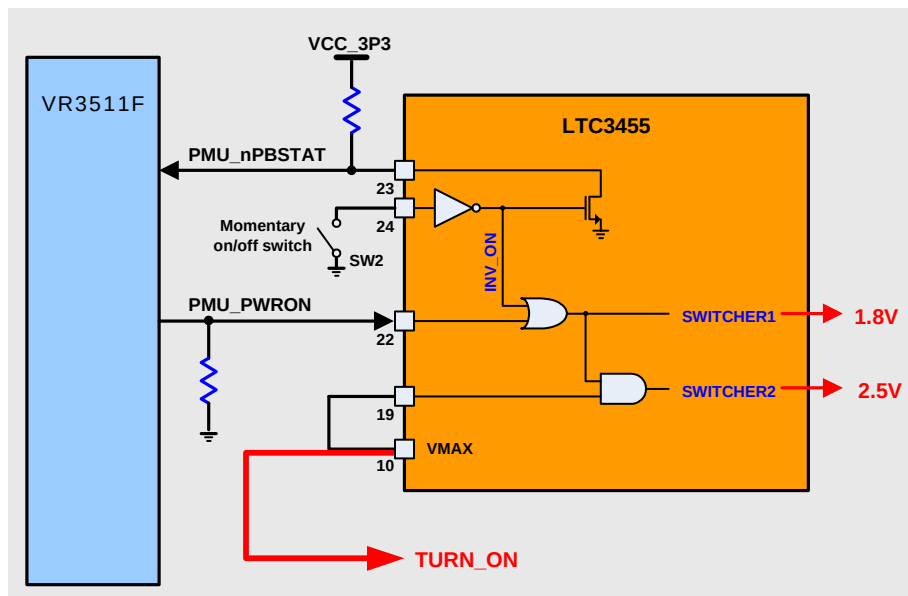
stable 5V power, two Step-Down DC/DC Converters(LT3430 & LT1765-5) are used. LTC3455 plays a major role as a power management unit. It includes two Step-Down DC/DC Converters, Battery charger, Reset generator, and Momentary-On/Off switch function. TEA1211HN is a Step-Up/Down DC/DC Converter for generating constant 3.3V power. LTC3200ES6-5 is a Step-Up DC/DC Converter. LT1932 is a White LED Driver IC for LCD back light.

3.2.1. Power Management Unit (LTC3455)

When only battery power is available, the only method to turn on the power is pressing the momentary on/off switch. When external adaptor power is applied, the LTC3455 seamlessly transitions from battery power to the external adaptor power. If the external adaptor power is applied, the LTC3455 is always turned on and supplies the power.

3.2.2. Operation of Momentary-On/Off Switch

<Fig 4> shows the method for using a momentary on/off switch to turn the LTC3455 on and off.



<Fig 4. Momentary on/off switch Operation>

POWER-ON Procedure

When the momentary on/off switch is first pressed, shorting the Pin24 to ground, PMU_nPBSTAT goes low and the LTC3455 first brings up the VMAX pin, then enables SWITCHER1 & SWITCHER2 to power the VR3511F. Once up and running, the VR3511F must provide HIGH-level PMU_PWRON signal to keep the LTC3455 turned on after the momentary

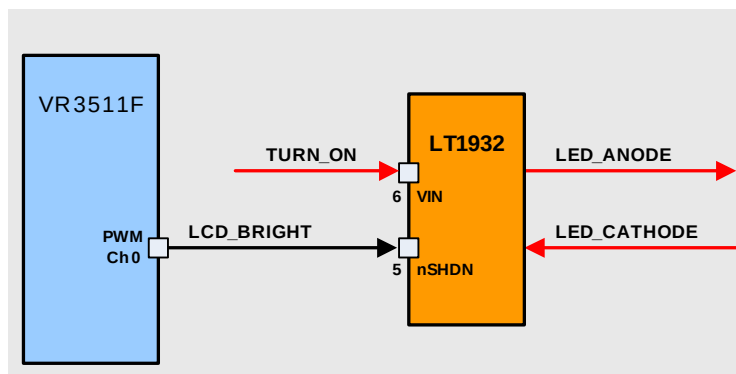
on/off switch is released.

POWER-OFF Procedure

When the momentary on/off switch is pressed again to turn off the power, the PMU_nPBSTAT signal is pulled low to notify the VR3511F that the momentary on/off switch has been pressed. The VR3511F prepares for shutdown then pulls the PMU_PWRON signal low. When the momentary on/off switch is released, the INV_ON signal goes low and SWITCHER1 & SWITCHER2 are disabled.

3.2.3. LCD Back light Power control

LT1932, the White LED Driver IC is used for LCD back light. It can drive Eight White LEDs from 3V. The brightness of LCD back light is controlled by PWM channel 0 of VR3511F.



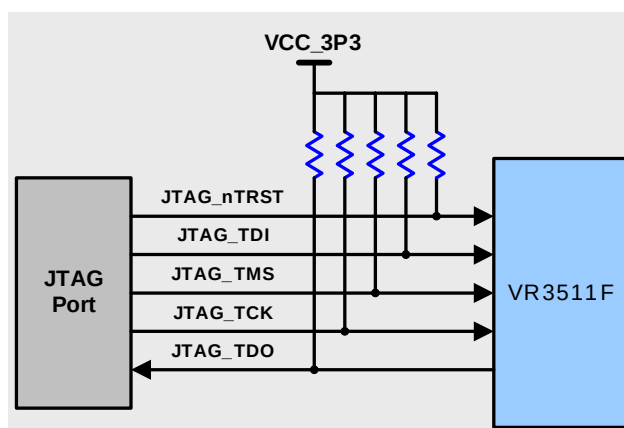
<Fig 5. Block Diagram of LCD Back light Power>

3.2.4. 3.3V Power generation

To generate the constant 3.3V power, the TEA1211HN DC/DC converter is used. It is a step up/down converter with high efficiency & large current driving capacity.

3.3. JTAG Interface

<Fig 6> shows the block diagram of JTAG interface. User can connect ARM Multi-ICE or J-flash connector to JTAG port. All JTAG signals must be connected to VCC via pull-up resistor.



<Fig 6. Block diagram of JTAG interface>

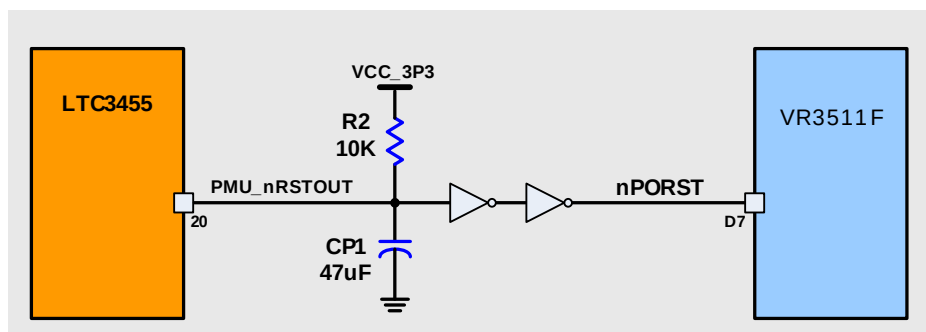
Signal Name	Type	Description
JTAG_nTRST	I	nTRST(TAP Controller Reset) resets the TAP controller at start.(10K Ω Pull-up register)
JTAG_TDI	I	TDI(TAP Controller Data Input) is the serial input for test instructions and data.(10K Ω pull-up register)
JTAG_TMS	I	TMS(TAP Controller Mode Select) controls sequence of the TAP Controller's states.(10K Ω pull-up register)
JTAG_TCK	I	TCK(TAP Controller Clock) provides the clock input for the JTAG logic.(10K Ω Pull-up register)
JTAG_TDO	O	TDO(TAP Controller Data Output) is the serial output for test instructions and data. (10K Ω Pull-up register)

<Table 2. Signal description of JTAG interface>

3.4. Reset

LTC3455 outputs the PMU_nRSTOUT reset signal (this reset pin is an open-drain output type, so external pull-up resistor is necessary). LTC3455 provides a 200ms reset signal during power-up, but for making more stable reset period of VR3511F, one delay capacitor(CP1) is necessary.

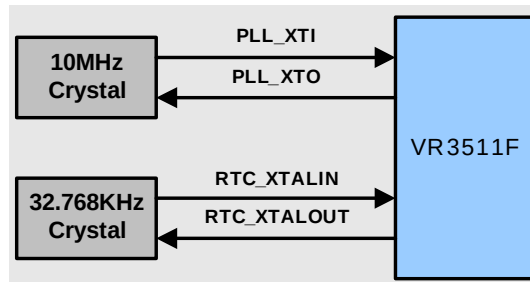
<Fig 7> shows the reset part of VR3511F Development Board 2.0.0.



<Fig 7. Block Diagram of Reset part>

3.5. Clock

10MHz and 32.768KHz Crystal are basic clock source of the VR3511F. 10MHz clock signal is an input for internal PLL and 32.768KHz clock signal is an input for RTC and used to make control signals (Static Bus control signals, Reset output signal, and so on).



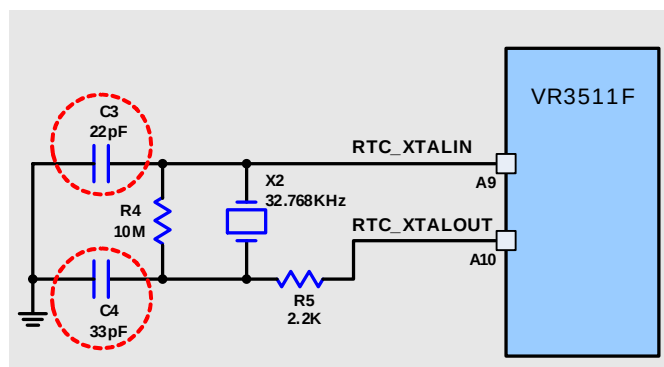
<Fig 8. Block diagram of Clock part>

Signal Name	Type	Description
PLL_XTI	I	Clock input for Internal PLL
PLL_XTO	O	Clock Output of Internal PLL
RTC_XTALIN	I	Clock input for RTC
RTC_XTALOUT	O	Clock Output for RTC

<Table 3. Signal description of Clock part>

Caution 2) Load capacitor selection of RTC Crystal

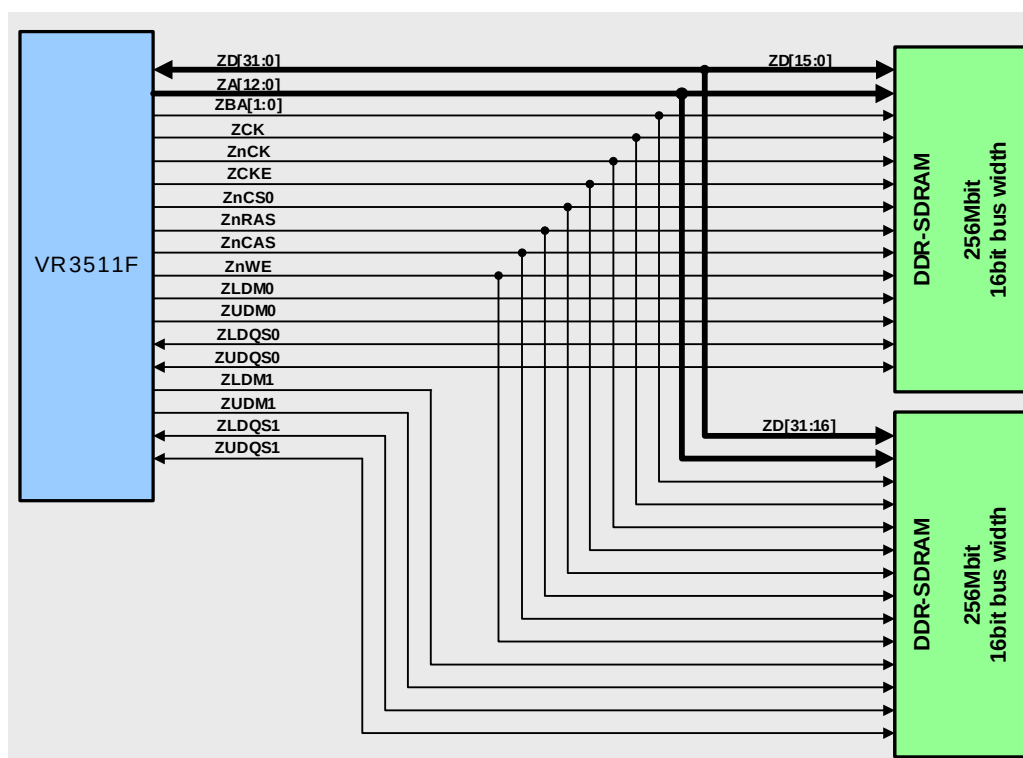
The customer of VR3511F has to pay attention to the load capacitor selection of RTC Crystal. <Fig 9> shows the detail circuit of RTC clock part of VR3511F Development Board 2.0.0. C3 and C4 value are 22pF and 33pF respectively. However it is not an absolute value because load capacitor value is dependent on Crystal vendor. 22pF & 33pF are just a value for VR3511F Development Board 2.0.0 case. Do not think these values are adaptable to your own VR3511F product. For stable oscillation of RTC Crystal(32.768KHz), H/W developer must check the load capacitor value that Crystal vendor recommend.



<Fig 9. Load capacitor selection for RTC Crystal>

3.6. DDR-SDRAM Memory Interface

<Fig 10> shows the block diagram of DDR-SDRAM memory interface. VR3511F has three DDR-SDRAM memory bank (X-bank, Y-bank, and Z-bank). For VR3511F Development Board 2.0.0 case, only the Z-bank is used as DDR-SDRAM memory bank. The total data bus width of Z-bank is 32 bit, so two DDR-SDRAM(with 16bit data bus width) is connected to Z-bank. The total DDR-SDRAM Memory capacity is 512Mbit (256Mbit x 2)



<Fig 10. Block diagram of DDR-SDRAM Memory Bank>

Signal Name	Type	Description
ZD[31:0]	I/O	Z-BUS DDR-SDRAM Data [31:0]
ZA[12:0]	O	Z-BUS DDR-SDRAM Address [12:0]
ZBA0	O	Z-BUS DDR-SDRAM Bank Address 0
ZBA1	O	Z-BUS DDR-SDRAM Bank Address 1
ZCK	O	Z-BUS DDR-SDRAM Clock
ZnCK	O	Z-BUS DDR-SDRAM Inverting Clock
ZCKE	O	Z-BUS DDR-SDRAM Clock Enable
ZnCS0	O	Z-BUS DDR-SDRAM Chip Select
ZnRAS	O	Z-BUS DDR-SDRAM RAS
ZnCAS	O	Z-BUS DDR-SDRAM CAS
ZnWE	O	Z-BUS DDR-SDRAM Write Enable
ZLDM0	O	Z-BUS DDR-SDRAM Data Mask for ZD[7:0]
ZUDM0	O	Z-BUS DDR-SDRAM Data Mask for ZD[15:8]
ZLDQS0	I/O	Z-BUS DDR-SDRAM Data Strobe for ZD[7:0]
ZUDQS0	I/O	Z-BUS DDR-SDRAM Data Strobe for ZD[15:8]

ZLDM1	O	Z-BUS DDR-SDRAM Data Mask for ZD[23:16]
ZUDM1	O	Z-BUS DDR-SDRAM Data Mask for ZD[31:24]
ZLDQS1	I/O	Z-BUS DDR-SDRAM Data Strobe for ZD[23:16]
ZUDQS1	I/O	Z-BUS DDR-SDRAM Data Strobe for ZD[31:24]

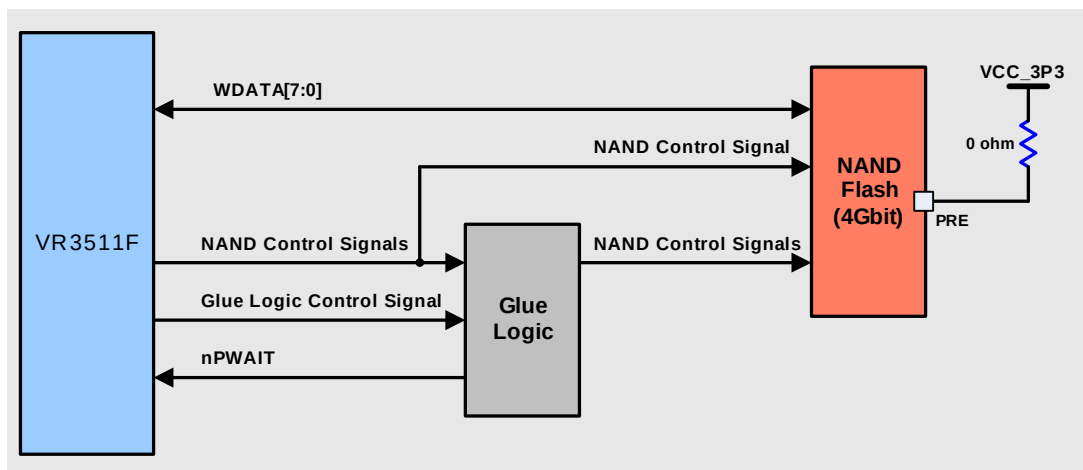
<Table 4. Signal description of DDR-SDRAM interface>

3.7. NAND Flash Interface

Basically the VR3511F does not support the boot from Large Block Size NAND Flash (only supports Small Block Size NAND boot). However taking advantage of PRE mode of Large Block Size NAND Flash, it is possible to boot from Large Block Size NAND Flash. VR3511F Development Board 2.0.0 supports Large Block Size NAND boot as well as Small Block Size NAND boot. The basic boot mode of VR3511F Development Board 2.0.0 is designed for Large Block NAND boot case and if a customer wants to use Small Block NAND boot, some resistors are must be changed according to below rules (of course you must replace Large Block NAND Flash with Small Block NAND Flash).

Change point for Small Block Size NAND Flash

- Do Not Install : R175, R184, R185, R277, R284
- Install : R176, R177, R186, R278



<Fig 11. Block Diagram of NAND Interface>

Signal Name	Type	Description
WDATA[7:0]	I/O	W-Bus Data [7:0]
WnNFOE	O	NAND Read Enable
nNFCS0_L	O	NAND Chip Enable for Large Block NAND Flash
WnNFCS0	O	NAND Chip Enable for Small Block NAND Flash
WnPIOW/NF_ALE	O	NAND Address Latch Enable
WnPIOR/NF_CLE	O	NAND Command Latch Enable

WnNFWE	O	NAND Write Enable
NAND_BOOT	O	Control signal for NAND Boot
PLL_XTO	O	Pseudo RnB signal for Large Block NAND Boot
RnB	I	Ready nBusy signal for Small Block NAND Boot
nPWAIT	I	Wait Input signal of External Device

<Table 5. Signal description of NAND interface>

3.8. LCD Interface

The LCD module of VR3511F Development Board 2.0.0 is LTG430WQ-F02 of SAMSUNG. The resolution and size of LCD module are WQVGA(480x272) and 4.3 inch respectively. LTG430WQ-F02 has serial-type 7 White LEDs for LCD backlight and the LED current is supplied from LT1932 White LED Driver IC.

RGB data bit order

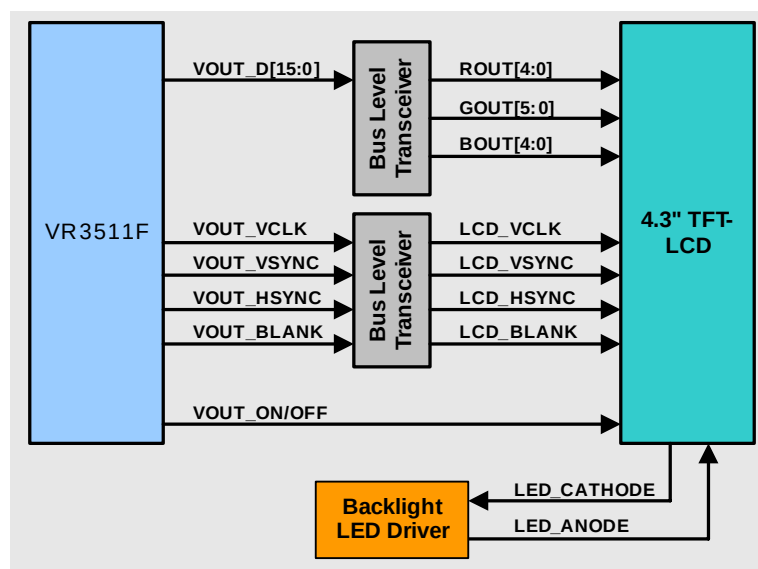
VOUT_D [15:11] → R [4:0]

VOUT_D [10:5] → G [5:0]

VOUT_D [4:0] → B [4:0]

Caution 3) LCD Interface

The input video signal level of LTG430WQ-F02 is not 3.3V but 2.5V, so Bus Level Transceiver IC(74VCX163245MTD) must be located between VR3511F and LTG430WQ-F02.



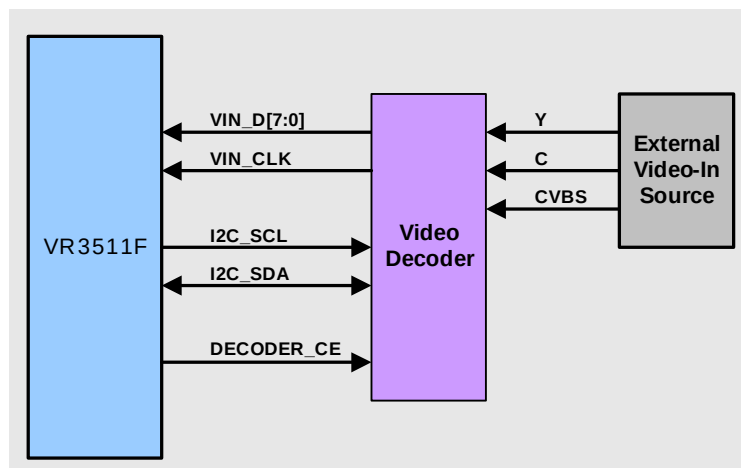
<Fig 12. Block Diagram of LCD Interface>

Signal Name	Type	Description
VOUT_D[15:0]	O	Digital Video Data Output[15:0]
VOUT_VCLK	O	Video Clock Output
VOUT_HSYNC	O	Video Horizontal Sync
VOUT_VSYNC	O	Video Vertical Sync
VOUT_BLANK	O	Video Data Enable
VOUT_ON/OFF	O	LCD On/Off signal
LED_CATHODE	-	CATHODE signal for White LED
LED_ANODE	-	ANODE signal for White LED

<Table 6. Signal description of LCD interface>

3.9. Video-In Interface

The Video Decoder in VR3511F Development Board 2.0.0 receives CVBS, Y/C analog video signal and outputs ITU656 YUV4:2:2 format video data signals to VR3511F.



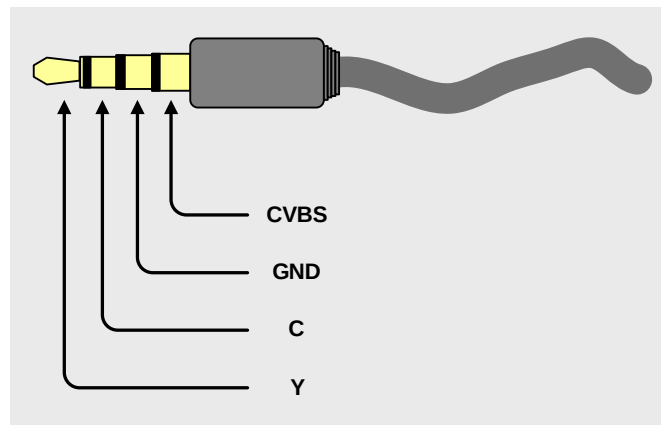
<Fig 13. Block Diagram of Video-In Interface>

Signal Name	Type	Description
VIN_D[7:0]	I	Video In Data Bus [7:0]
VIN_CLK	I	Video In Clock
I2C_SCL	O	I2C Clock signal
I2C_SDA	I/O	I2C Data signal
DECODER_CE	O	Video Decoder(SAA7113) Enable signal

<Table 7. Signal description of Video-In interface>

Caution 4) Pin assignment of Video-In Cable

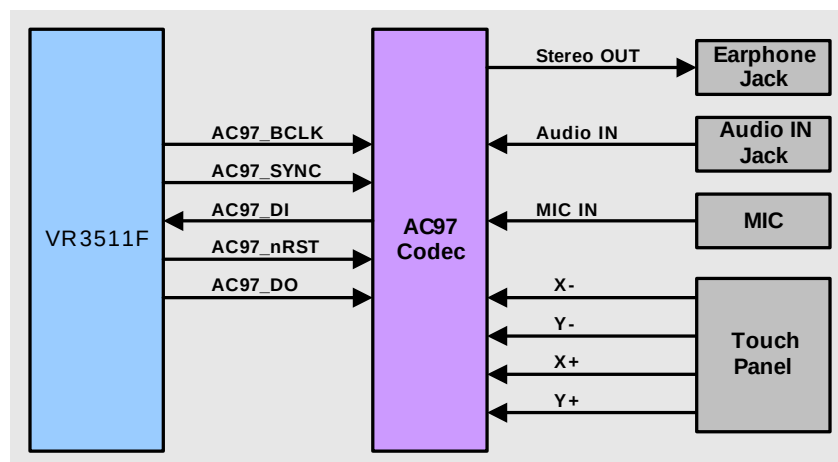
Below <Fig 14> shows the pin assignment of Video-In Cable for VR3511F Development Board 2.0.0. Keep it in mind that Video-In cable is not included in VR3511F DTK package.



<Fig 14. Pin assignment of Video-In Cable for VR3511F Development Board 2.0.0>

3.10. AC97 Codec Interface

The AC97 Codec IC(WM9712) in VR3511F Development Board 2.0.0 has Audio Out, Audio In, and MIC In function and includes touch controller. Touch data signals(X+, X-, Y+, Y-) from LCD Touch Panel is connected to WM9712.



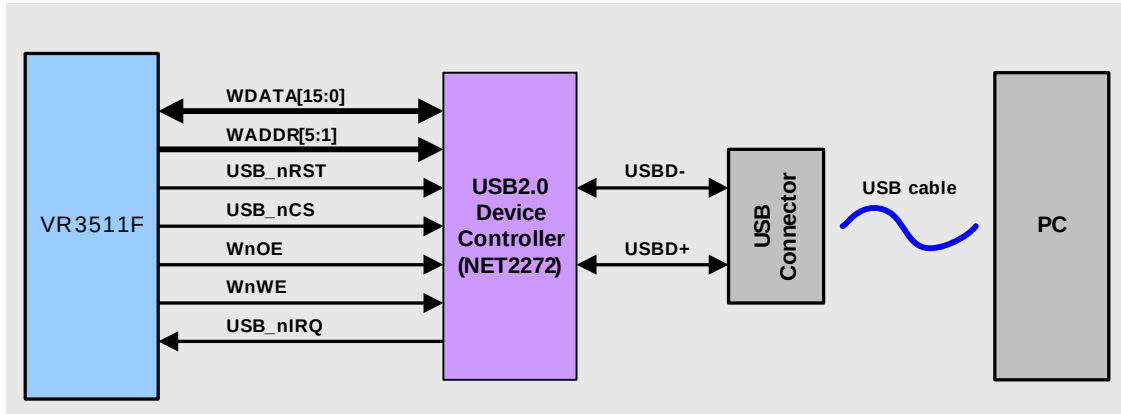
<Fig 15. Block Diagram of AC97 Codec Interface>

Signal Name	Type	Description
AC97_BCLK	O	AC97 Clock Output
AC97_SYNC	O	AC97 Sync
AC97_DI	I	AC97 Data In
AC97_nRST	O	AC97 Reset
AC97_DO	O	AC97 Data Output
X-	-	X- signal from Touch Panel
Y-	-	Y- signal from Touch Panel
X+	-	X+ signal from Touch Panel
Y+	-	Y+ signal from Touch Panel

<Table 8. Signal description of AC97 Codec interface>

3.11. USB2.0 Controller Interface

<Fig 16> shows the block diagram of USB2.0 Device Controller Interface.



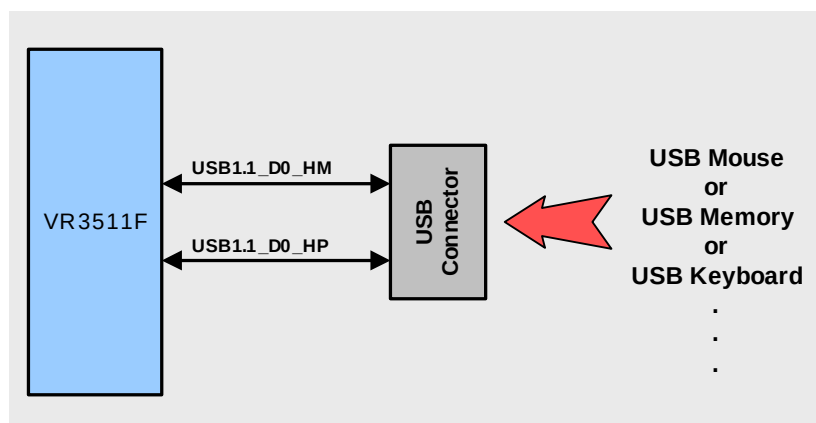
<Fig 16. Block Diagram of USB2.0 part>

Signal Name	Type	Description
WDATA[15:0]	I/O	W-BUS Data[15:0]
WADDR[5:1]	O	W-BUS Address[5:1]
USB_nRST	O	USB device controller Reset signal
USB_nCS	O	USB device controller Chip Select signal
WnOE	O	USB device controller Read Enable signal
WnWE	O	USB device controller Write Enable signal
USB_nIRQ	I	

<Table 9. Signal description of USB2.0 Device Controller interface>

3.12. USB1.1 Host Interface

<Fig 17> shows the block diagram of USB1.1 Host Interface. User can connect & use USB Mouse or USB MemoryStick or USB Keyboard and so on.



<Fig 17. Block Diagram of USB1.1 Host Interface>

Signal Name	Type	Description
USB1.1_D0_HM	I/O	USB1.1 Host Data Minus
USB1.1_D0_HP	I/O	USB1.1 Host Data Plus

<Table 10. Signal description of USB1.1 Host interface>

3.13. USB1.1 Device Interface

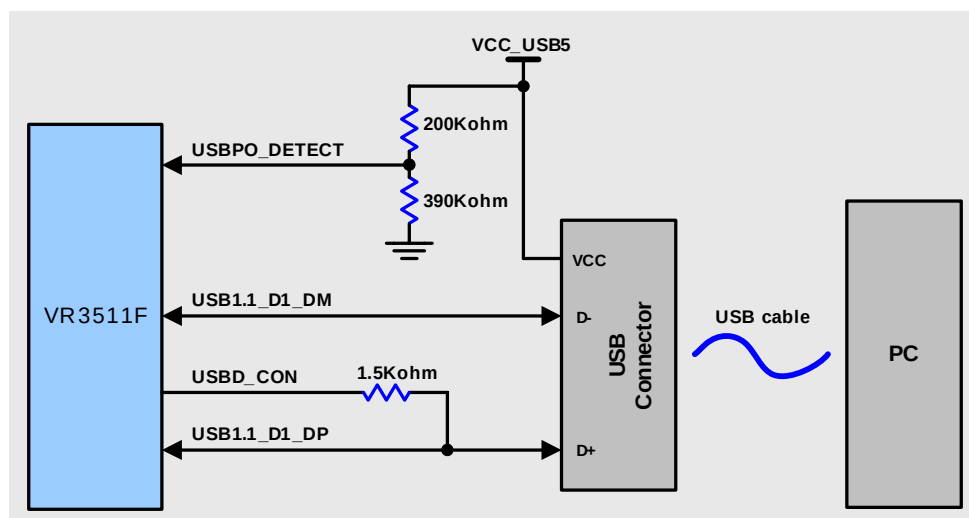
<Fig 18> shows the detail design about USB1.1 Device Interface. Two GPIO signals are used for PC detection and USBD+ signal control.

USBPO_DETECT

PC(USB1.1 Host) detection signal. This signal is used to detect whether PC is connected or not. Normal status of USBPO_DETECT is LOW level. If PC is connected to USB1.1 Device port of VR3511F Development Board 2.0.0, this signal's value is changed from LOW level to HIGH level.

USB_D_CON

USB Data Plus control signal. This signal is used to control the Pull-up & Pull-down of Data Plus signal of USB1.1 Device. If PC is not connected to USB1.1 Device port of VR3511F Development Board 2.0.0, USB_D_CON must maintain LOW level by Software Program. If PC is connected(VR3511F can recognize the PC connection by reading the USBPO_DETECT signal's value), Software Program initializes the USB1.1 Device Driver of VR3511F and then must change USB_D_CON signal's value from LOW to HIGH to inform PC of external USB1.1 Device connection.



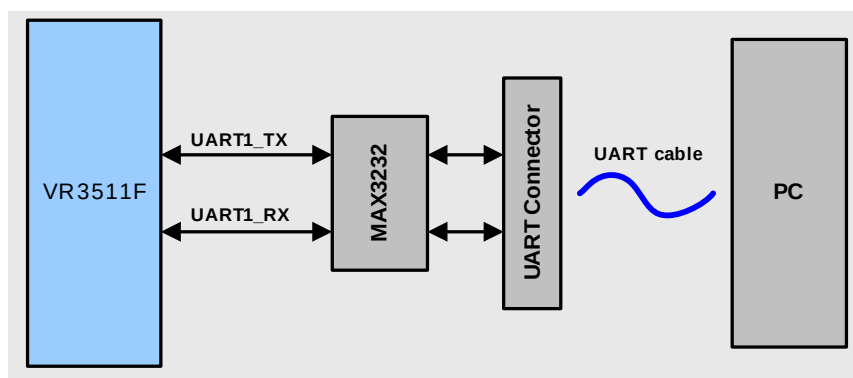
<Fig 18. Block Diagram of USB1.1 Device Interface>

Signal Name	Type	Description
USB1.1_D1_DM	I/O	USB1.1 Device Data Minus
USB1.1_D1_DP	I/O	USB1.1 Device Data Plus
USB_D_CON	O	USB1.1 Device Data Plus Pull-Up Control signal
USBPO_DETECT	I	USB1.1 Device Detection signal

<Table 11. Signal description of USB1.1 Device interface>

3.14. UART Interface

VR3511F has two UART channels(ch0, ch1) and channel 1 is allocated to PC debugging port. <Fig 19> shows the block diagram of UART channel 1 PC debugging port. UART channel 0 is allocated to GPS connection port (**GPS module is not included in VR3511F DTK Package**).



<Fig 19. Block Diagram of UART Debugging Port>

Signal Name	Type	Description
UART1_RX	I	UART Input signal from PC
UART1_TX	O	UART Output signal to PC

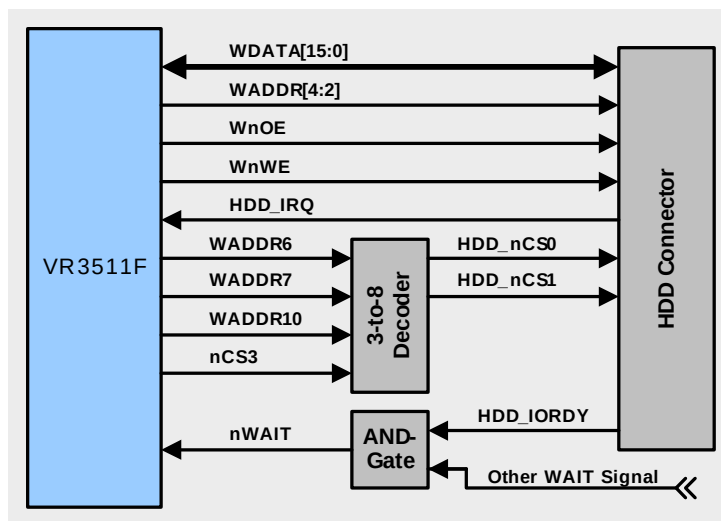
<Table 12. Signal description of UART channel 1 interface>

3.15. IDE Interface

Originally, HDD is connected to IDE controller of VR3511F. However in this case, only PIO mode is usable, so there is a restriction of data-transfer performance. **Therefore MagicEyes strongly recommend the workaround solution(which is called Static IDE) to customer.** In Static IDE case, HDD is connected to Static Controller of VR3511F and the data transfer between HDD and DDR-SDRAM is achieved by memory to memory DMA(not Multi-word DMA or Ultra DMA), and as a result, the data transfer performance of Static IDE case is increased than IDE controller case. The IDE interface of VR3511F Development Board 2.0.0 is designed as Static IDE.

Caution 5) Supported IDE device driver

Keep it mind that MagicEyes only provides IDE device driver for Static controller (we do not provide & support IDE device driver for IDE controller).



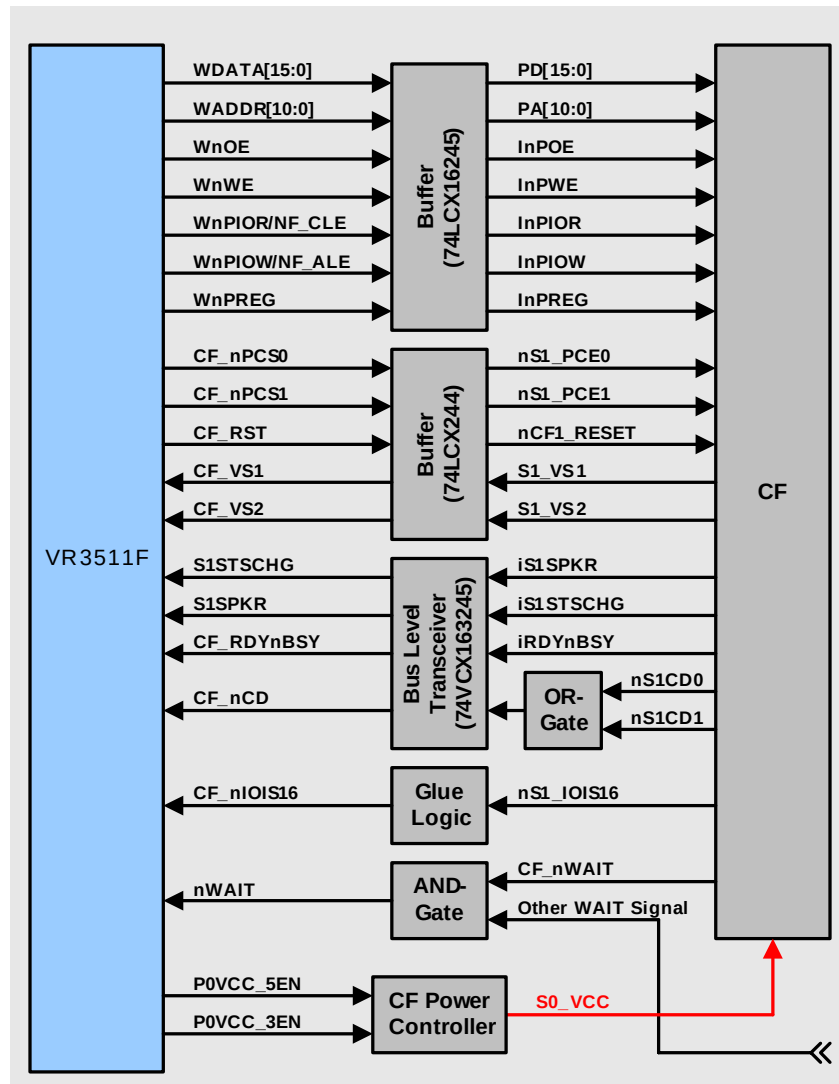
<Fig 20. Block Diagram of IDE Interface>

Signal Name	Type	Description
WDATA[15:0]	I/O	W-BUS Data[15:0]
WADDR[4:2]	O	W-BUS Address[4:2]
WnOE	O	W-BUS Read Enable signal
WnWE	O	W-BUS Write Enable signal
HDD_IRQ	I	HDD Interrupt Request
HDD_nCS0	-	HDD Chip Select signal 0. This signal is made by address decoding method
HDD_nCS1	-	HDD Chip Select signal 1. This signal is made by address decoding method
nCS3	O	Static bus Chip Select signal 3. This signal is used for making the HDD Chip Select signal
HDD_IORDY		HDD Wait signal

<Table 13. Signal description of IDE interface>

3.16. CF Interface

<Fig 21> shows the block diagram of CF interface.



<Fig 21. Block Diagram of CF Interface>

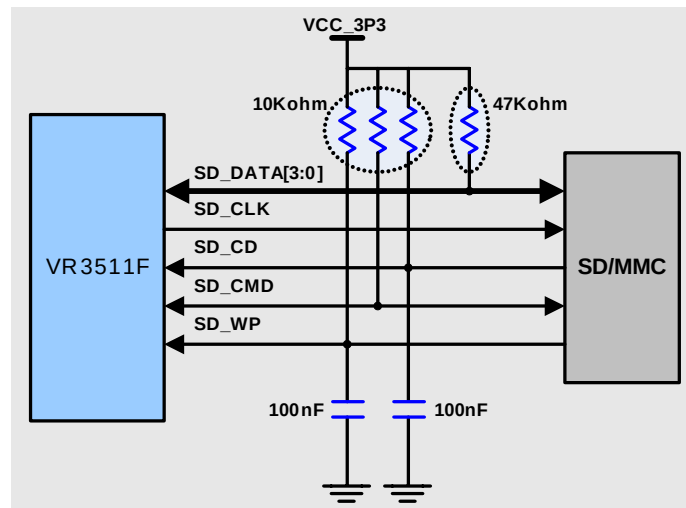
Signal Name	Type	Description
WDATA[15:0]	I/O	W-BUS Data[15:0]
WADDR[10:0]	O	W-BUS Address[10:0]
WnOE	O	PCMCIA Memory Read Enable
WnWE	O	PCMCIA Memory Write Enable
WnPior/NF_CLE	O	PCMCIA IO Read Enable
WnPior/NF_ALE	O	PCMCIA IO Write Enable
WnPREG	O	This signal is an output and indicates that, on a memory transaction, the target address is attribute space. This signal has same timing as address
CF_nIOIS16	I	It is an input and is an acknowledgement from the PCMCIA card that the current address is a valid 16bit wide I/O address
CF_nPCS0	O	PCMCIA Chip Select 0
CF_nPCS1	O	PCMCIA Chip Select 1
CF_nCD	I	PCMCIA Card Detect signal
CF_VS1	I	PCMCIA Voltage Sense 1
CF_VS2	I	PCMCIA Voltage Sense 2

S1STSCHG	I	This signal has a meaning for PC Card IO Mode case. It is asserted low to alert the host to changes in the READY and Write Protect states, while the IO interface is configured. Its use is controlled by the Card Config and Status Register.
S1SPKR	I	This signal has a meaning for PC Card IO Mode case. It is the Binary Audio output from the card. If the Card does not support the Binary Audio function, it should be held negated.
CF_RDYnBSY	I	PCMCIA Ready nBusy signal
CF_nWAIT	I	PCMCIA Wait signal
CF_RST	O	PCMCIA Reset signal

<Table 14. Signal description of CF interface>

3.17. SD/MMC Interface

<Fig 22> shows the block diagram of SD/MMC interface.



<Fig 22. Block Diagram of SD/MMC Interface>

Caution 6) Pull-up resistor value of SD/MMC signals

Below values are pull-up resistor value of SD/MMC signals.

SD_DATA[3:0] : 47Kohm
SD_CLK : None
SD_CD : 10Kohm
SD_WP : 10Kohm
SD_CMD : 10Kohm or 4.7Kohm

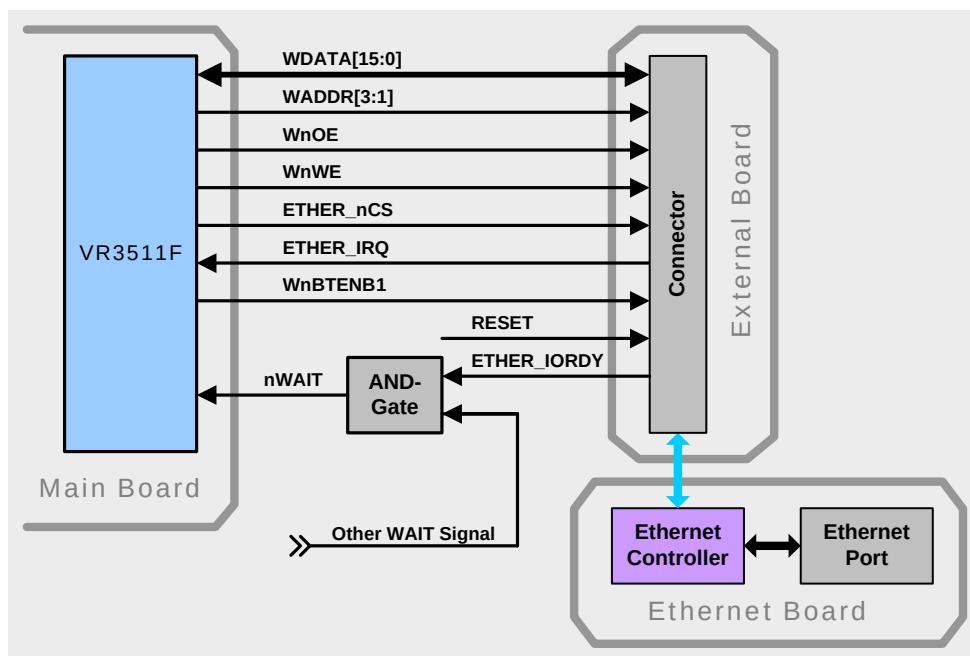
CMD pin of MMC card is open-drain output type, so the rising transition time of CMD signal is dependent on external pull-up resistor value(therefore too large resistor value causes the distortion of CMD signal). MagicEyes recommends 10Kohm or 4.7Kohm to customer.

Signal Name	Type	Description
SD_DATA[3:0]	I/O	SD/MMC Data[3:0]
SD_CLK	O	SD/MMC Clock
SD_CD	I	SD/MMC Card Detect signal
SD_CMD	I/O	SD/MMC Command
SD_WP	I	SD/MMC Write Protect

<Table 15. Signal description of SD/MMC interface>

3.18. Ethernet Controller Interface

Ethernet sub board is connected to External board and External board is connected to VR3511F Development Board 2.0.0. <Fig 23> shows the concept of board connection.



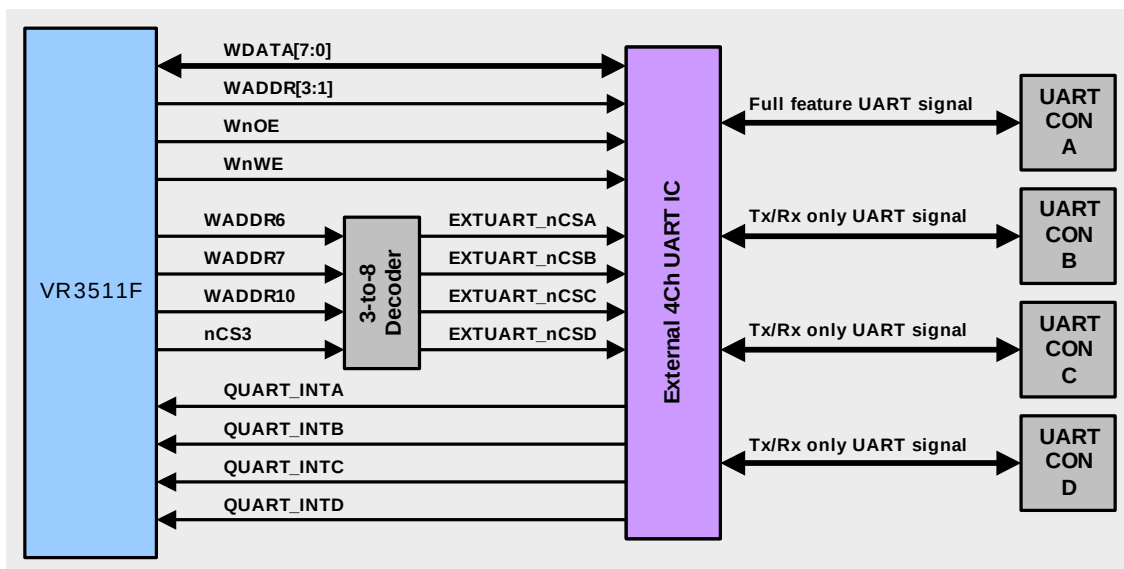
<Fig 23. Block Diagram of Ethernet Controller Interface>

Signal Name	Type	Description
WDATA[15:0]	I/O	W-BUS Data[15:0]
WADDR[3:1]	O	W-BUS Address[3:1]
WnOE	O	W-BUS Read Enable signal
WnWE	O	W-BUS Write Enable signal
ETHER_nCS	O	Ethernet controller Chip Select
ETHER_IRQ	I	Ethernet controller Interrupt Request
ETHER_IORDY	I	Ethernet controller Wait signal
WnBTENB1	O	High Byte Enable signal
RESET	-	Ethernet controller Reset signal

<Table 16. Signal description of Ethernet Controller interface>

3.19. External UART IC Interface

VR3511F Development Board 2.0.0 has six channel UART ports. Two of them is directly supported by VR3511F and the rest 4 ports are supported by External 4 channel UART IC. External UART IC is mounted in External Board. It supports full-featured UART signal x 1 channel and Tx/Rx UART signal x 3 channels.



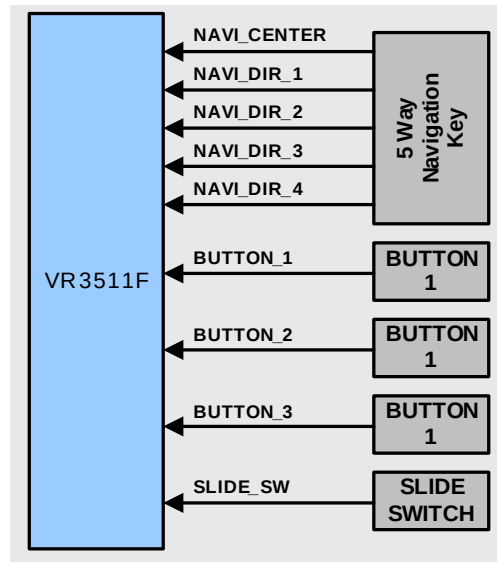
<Fig 24. Block Diagram of External UART Controller Interface>

Signal Name	Type	Description
WDATA[7:0]	I/O	W-BUS Data[7:0]
WADDR[3:1]	O	W-BUS Address[3:1]
WnOE	O	W-BUS Read Enable signal
WnWE	O	W-BUS Write Enable signal
EXTUART_nCSA	-	External UART IC Chip Select signal A. This signal is made by address decoding method
EXTUART_nCSB	-	External UART IC Chip Select signal B. This signal is made by address decoding method
EXTUART_nCSC	-	External UART IC Chip Select signal C. This signal is made by address decoding method
EXTUART_nCSD	-	External UART IC Chip Select signal D. This signal is made by address decoding method
nCS3	O	Static bus Chip Select signal 3. This signal is used for making the External UART Chip Select signal
QUART_INTA	I	External UART IC Interrupt Request A
QUART_INTB	I	External UART IC Interrupt Request B
QUART_INTC	I	External UART IC Interrupt Request C
QUART_INTD	I	External UART IC Interrupt Request D

<Table 17. Signal description of External UART Controller interface>

3.20. Key Function

VR3511F Development Board 2.0.0 has one Navigation Key(5-direction), three Push button, and one Slide switch.



<Fig 25. Block Diagram of Key Function>

Signal Name	Type	Description
NAVI_CENTER	I	Center signal of Navigation Key
NAVI_DIR_1	I	Direction 1 signal of Navigation Key
NAVI_DIR_2	I	Direction 2 signal of Navigation Key
NAVI_DIR_3	I	Direction 3 signal of Navigation Key
NAVI_DIR_4	I	Direction 4 signal of Navigation Key
BUTTON_1	I	Push Button input signal of BUTTON 1
BUTTON_1	I	Push Button input signal of BUTTON 1
BUTTON_1	I	Push Button input signal of BUTTON 1
SLIDE_SW	I	Switch input signal of SLIDE SWITCH

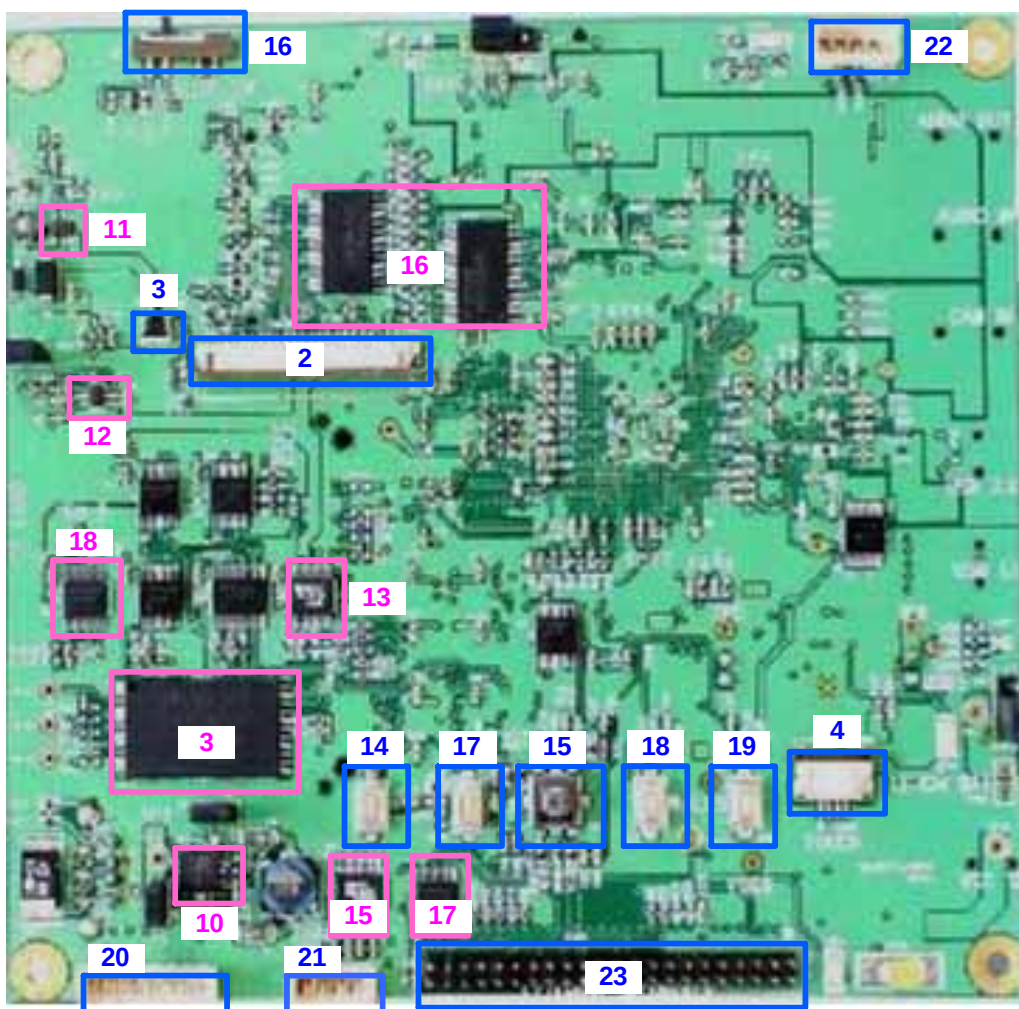
<Table 18. Signal description of Key Function>

4. Components Description

4.1. Components Arrangement

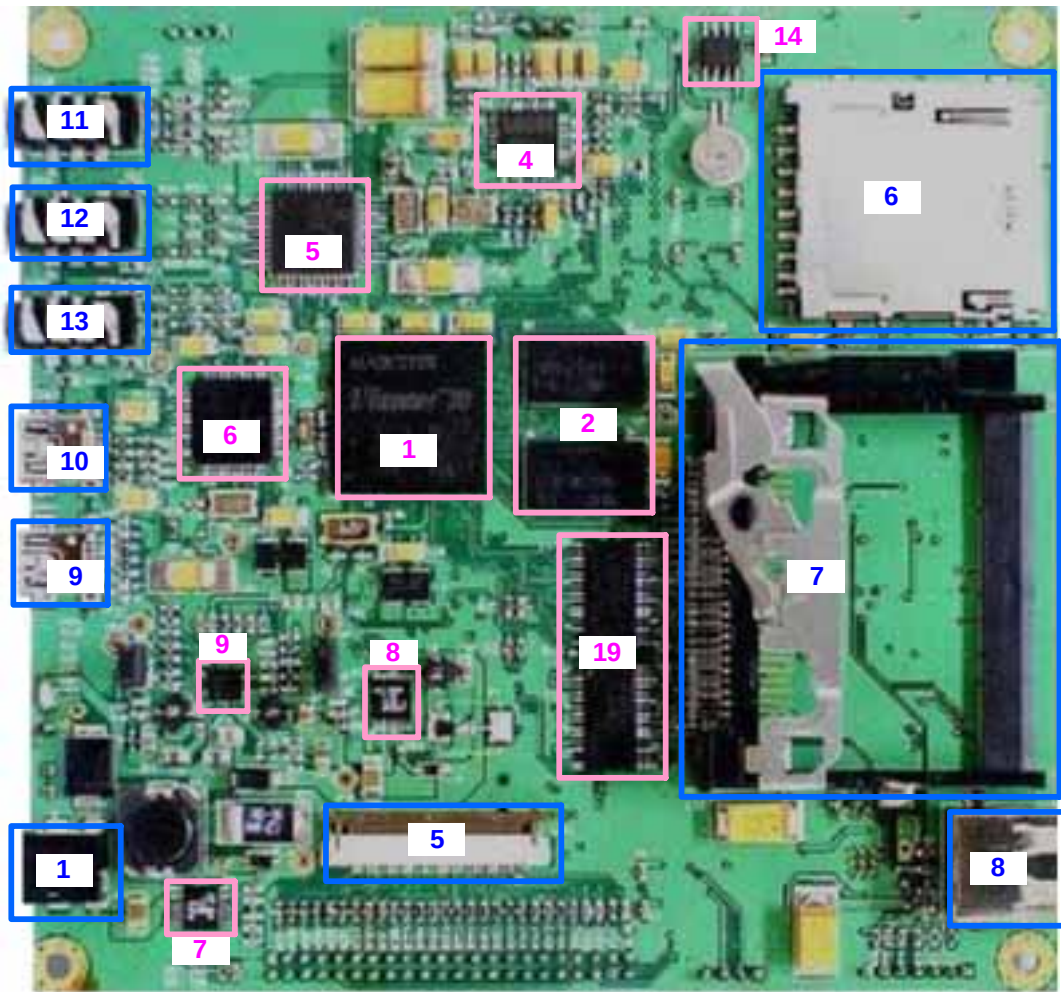
Below <Fig 26> and <Fig 27> show the arrangement of major components in VR3511F Development Board 2.0.0. Red-colored components mean IC and blue-colored components mean connector & switch.

4.1.1. Top View



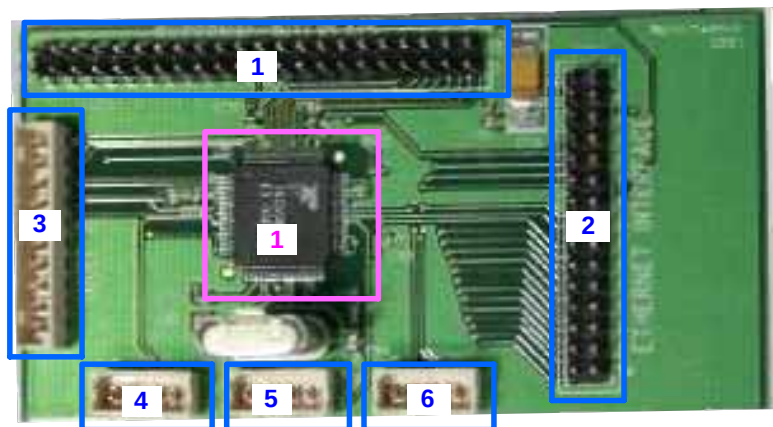
<Fig 26. Top View of VR3511F Development Board 2.0.0>

4.1.2. Bottom View



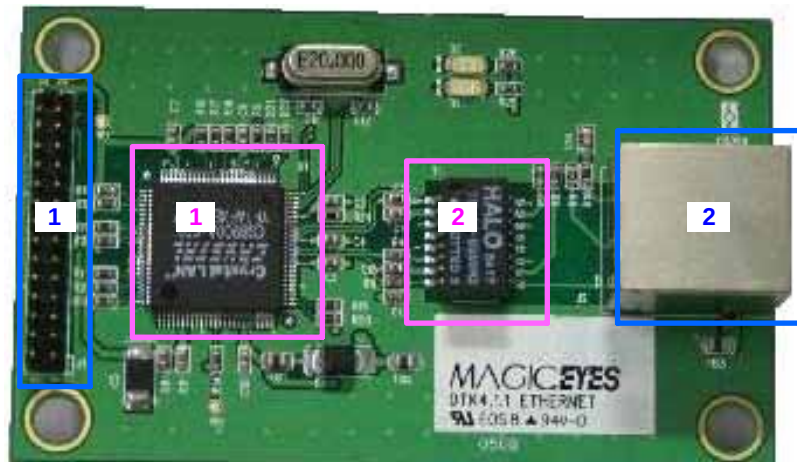
<Fig 27. Bottom View of VR3511F Development Board 2.0.0>

4.1.3. External Board



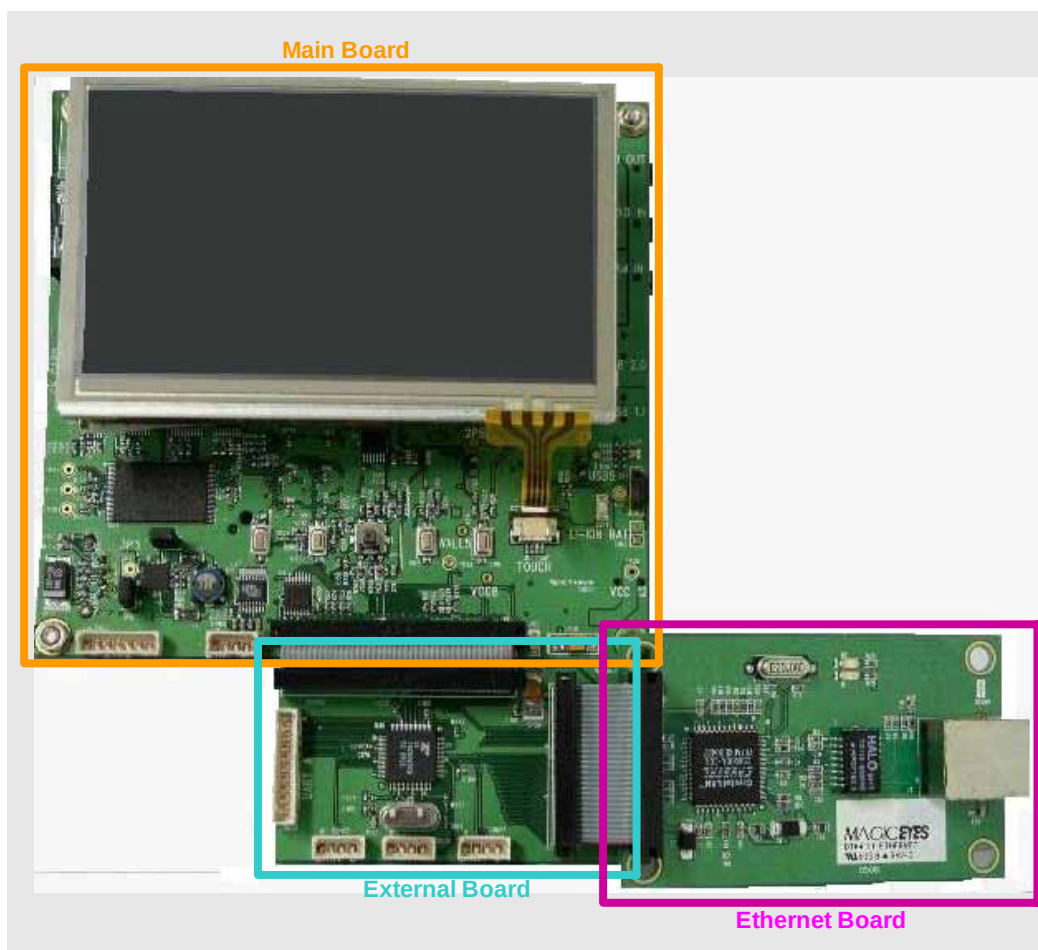
<Fig 28. Components arrangement of External Board>

4.1.4. Ethernet Board



<Fig 29. Components arrangement of External Board>

4.1.5. Board to board connection



<Fig 30. Board to board connection>

4.2. Components Description

4.2.1. Main Board

- IC components

No	Ref No	Part Name	Description
1	U1	VR3511F	Main Processor Vendor : MagicEyes
2	U6, U7	K4H561638F-GCB3	DDR-SDRAM Capacity : 32MB Data bit width : 16bit Vendor : Samsung
3	U23	K9K4G08U0M	NAND Flash Capacity : 4Gbit Data bit width : 8bit Vendor : Samsung
4	U13	WM9712EFL	AC97 Codec with touch controller Vendor : Wolfson
5	U17	SAA7113H	Video Decoder Vendor : Philips
6	U18	NET2272	USB2.0 Device Controller Vendor : PLX
7	U9	LT3430	Step Down DC-DC converter Vendor : LT
8	U10	LT1765-5	Step Down DC-DC converter Vendor : LT
9	U8	LTC3455EUF	Power Management Unit Battery charger function Include two Step Down DC-DC converter Reset generator function Momentary on/off power switch function Battery charge current gauge function Vendor : LT
10	U12	TEA1211HN	Step Up/Down DC-DC converter Vendor : Philips
11	U11	LT1932	White LED Driver IC for LCD backlight Vendor : LT
12	U15	LT3200ES6-5	Step Up DC-DC converter Vendor : LT
13	U26	MIC2562A-1BTS	CF power controller No of channel : 1 Vendor : MICREL
14	U5	DS1672S-33	RTC IC Vendor : MAXIM
15	U20	MAX3232ECAE	UART Transceiver IC Vendor : MAXIM
16	U14, U16	74VCX163245MTD	Bus Level Transceiver Vendor : Fairchild
17	U4	74LCX138-MTC16	3 to 8 Decoder/Demultiplexer Vendor : Fairchild
18	U24	74LVC257APW	Quad 2-input multiplexer Vendor : Philips
19	U25, U27	74LCX16245	16bit bidirectional buffer Vendor : Fairchild

<Table 19. IC Components description of Main Board>

- Connector & Switch components

No	Ref No	Part Name	Description																																																																																		
1	CON4	TDC-018	12V Power adaptor Jack <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>VCC12</td></tr><tr><td>2</td><td>GND</td></tr></table>	Pin Description		1	VCC12	2	GND																																																																												
Pin Description																																																																																					
1	VCC12																																																																																				
2	GND																																																																																				
2	CON10	9631S_TOP	4.3"LCD connector. LCD : Samsung LTG430WQ-F02 <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>GND</td></tr><tr><td>2</td><td>GND</td></tr><tr><td>3</td><td>VCC_2P5</td></tr><tr><td>4</td><td>VCC_2P5</td></tr><tr><td>5</td><td>GND</td></tr><tr><td>6</td><td>GND</td></tr><tr><td>7</td><td>GND</td></tr><tr><td>8</td><td>ROUT0</td></tr><tr><td>9</td><td>ROUT1</td></tr><tr><td>10</td><td>ROUT2</td></tr><tr><td>11</td><td>ROUT3</td></tr><tr><td>12</td><td>ROUT4</td></tr><tr><td>13</td><td>GND</td></tr><tr><td>14</td><td>GND</td></tr><tr><td>15</td><td>GOUT0</td></tr><tr><td>16</td><td>GOUT1</td></tr><tr><td>17</td><td>GOUT2</td></tr><tr><td>18</td><td>GOUT3</td></tr><tr><td>19</td><td>GOUT4</td></tr><tr><td>20</td><td>GOUT5</td></tr><tr><td>21</td><td>GND</td></tr><tr><td>22</td><td>GND</td></tr><tr><td>23</td><td>GND</td></tr><tr><td>24</td><td>BOUT0</td></tr><tr><td>25</td><td>BOUT1</td></tr><tr><td>26</td><td>BOUT2</td></tr><tr><td>27</td><td>BOUT3</td></tr><tr><td>28</td><td>BOUT4</td></tr><tr><td>29</td><td>GND</td></tr><tr><td>30</td><td>LCD_VCLK</td></tr><tr><td>31</td><td>VOUT_ON/OFF</td></tr><tr><td>32</td><td>LCD_HSYNC</td></tr><tr><td>33</td><td>LCD_VSYNC</td></tr><tr><td>34</td><td>LCD_BLANK</td></tr><tr><td>35</td><td>VCC_LCD5</td></tr><tr><td>36</td><td>VCC_LCD5</td></tr><tr><td>37</td><td>N.C.</td></tr><tr><td>38</td><td>N.C.</td></tr><tr><td>39</td><td>GND</td></tr><tr><td>40</td><td>GND</td></tr></table>	Pin Description		1	GND	2	GND	3	VCC_2P5	4	VCC_2P5	5	GND	6	GND	7	GND	8	ROUT0	9	ROUT1	10	ROUT2	11	ROUT3	12	ROUT4	13	GND	14	GND	15	GOUT0	16	GOUT1	17	GOUT2	18	GOUT3	19	GOUT4	20	GOUT5	21	GND	22	GND	23	GND	24	BOUT0	25	BOUT1	26	BOUT2	27	BOUT3	28	BOUT4	29	GND	30	LCD_VCLK	31	VOUT_ON/OFF	32	LCD_HSYNC	33	LCD_VSYNC	34	LCD_BLANK	35	VCC_LCD5	36	VCC_LCD5	37	N.C.	38	N.C.	39	GND	40	GND
Pin Description																																																																																					
1	GND																																																																																				
2	GND																																																																																				
3	VCC_2P5																																																																																				
4	VCC_2P5																																																																																				
5	GND																																																																																				
6	GND																																																																																				
7	GND																																																																																				
8	ROUT0																																																																																				
9	ROUT1																																																																																				
10	ROUT2																																																																																				
11	ROUT3																																																																																				
12	ROUT4																																																																																				
13	GND																																																																																				
14	GND																																																																																				
15	GOUT0																																																																																				
16	GOUT1																																																																																				
17	GOUT2																																																																																				
18	GOUT3																																																																																				
19	GOUT4																																																																																				
20	GOUT5																																																																																				
21	GND																																																																																				
22	GND																																																																																				
23	GND																																																																																				
24	BOUT0																																																																																				
25	BOUT1																																																																																				
26	BOUT2																																																																																				
27	BOUT3																																																																																				
28	BOUT4																																																																																				
29	GND																																																																																				
30	LCD_VCLK																																																																																				
31	VOUT_ON/OFF																																																																																				
32	LCD_HSYNC																																																																																				
33	LCD_VSYNC																																																																																				
34	LCD_BLANK																																																																																				
35	VCC_LCD5																																																																																				
36	VCC_LCD5																																																																																				
37	N.C.																																																																																				
38	N.C.																																																																																				
39	GND																																																																																				
40	GND																																																																																				
3	CON11	04-6277-004-001-883	White LED driver connector <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>LED_CATHODE</td></tr><tr><td>2</td><td>N.C.</td></tr><tr><td>3</td><td>N.C.</td></tr><tr><td>4</td><td>LED_ANODE</td></tr></table>	Pin Description		1	LED_CATHODE	2	N.C.	3	N.C.	4	LED_ANODE																																																																								
Pin Description																																																																																					
1	LED_CATHODE																																																																																				
2	N.C.																																																																																				
3	N.C.																																																																																				
4	LED_ANODE																																																																																				
4	CON7	10019HR-04A00	Touch connector																																																																																		

			<table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>X-</td></tr><tr><td>2</td><td>Y-</td></tr><tr><td>3</td><td>X+</td></tr><tr><td>4</td><td>Y+</td></tr></table>	Pin Description		1	X-	2	Y-	3	X+	4	Y+																																																																								
Pin Description																																																																																					
1	X-																																																																																				
2	Y-																																																																																				
3	X+																																																																																				
4	Y+																																																																																				
5	CON21	FA5B040HP1	HDD connector <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>N.C.</td></tr><tr><td>2</td><td>N.C.</td></tr><tr><td>3</td><td>IDE_nRESET</td></tr><tr><td>4</td><td>GND</td></tr><tr><td>5</td><td>IDE_D7</td></tr><tr><td>6</td><td>IDE_D8</td></tr><tr><td>7</td><td>IDE_D6</td></tr><tr><td>8</td><td>IDE_D9</td></tr><tr><td>9</td><td>IDE_D5</td></tr><tr><td>10</td><td>IDE_D10</td></tr><tr><td>11</td><td>IDE_D4</td></tr><tr><td>12</td><td>IDE_D11</td></tr><tr><td>13</td><td>IDE_D3</td></tr><tr><td>14</td><td>IDE_D12</td></tr><tr><td>15</td><td>IDE_D2</td></tr><tr><td>16</td><td>IDE_D13</td></tr><tr><td>17</td><td>IDE_D1</td></tr><tr><td>18</td><td>IDE_D14</td></tr><tr><td>19</td><td>IDE_D0</td></tr><tr><td>20</td><td>IDE_D15</td></tr><tr><td>21</td><td>GND</td></tr><tr><td>22</td><td>N.C.</td></tr><tr><td>23</td><td>GND</td></tr><tr><td>24</td><td>IDE_nIOW</td></tr><tr><td>25</td><td>IDE_nIOR</td></tr><tr><td>26</td><td>GND</td></tr><tr><td>27</td><td>IDE_nIORDY</td></tr><tr><td>28</td><td>GND</td></tr><tr><td>29</td><td>N.C.</td></tr><tr><td>30</td><td>IDE_IRQ</td></tr><tr><td>31</td><td>IDE_ADDR1</td></tr><tr><td>32</td><td>TPOINT_1</td></tr><tr><td>33</td><td>IDE_ADDR0</td></tr><tr><td>34</td><td>IDE_ADDR2</td></tr><tr><td>35</td><td>IDE_NCS0</td></tr><tr><td>36</td><td>IDE_NCS1</td></tr><tr><td>37</td><td>DASP</td></tr><tr><td>38</td><td>VCC_3P3</td></tr><tr><td>39</td><td>VCC_3P3</td></tr><tr><td>40</td><td>IDE_DEVADR</td></tr></table>	Pin Description		1	N.C.	2	N.C.	3	IDE_nRESET	4	GND	5	IDE_D7	6	IDE_D8	7	IDE_D6	8	IDE_D9	9	IDE_D5	10	IDE_D10	11	IDE_D4	12	IDE_D11	13	IDE_D3	14	IDE_D12	15	IDE_D2	16	IDE_D13	17	IDE_D1	18	IDE_D14	19	IDE_D0	20	IDE_D15	21	GND	22	N.C.	23	GND	24	IDE_nIOW	25	IDE_nIOR	26	GND	27	IDE_nIORDY	28	GND	29	N.C.	30	IDE_IRQ	31	IDE_ADDR1	32	TPOINT_1	33	IDE_ADDR0	34	IDE_ADDR2	35	IDE_NCS0	36	IDE_NCS1	37	DASP	38	VCC_3P3	39	VCC_3P3	40	IDE_DEVADR
Pin Description																																																																																					
1	N.C.																																																																																				
2	N.C.																																																																																				
3	IDE_nRESET																																																																																				
4	GND																																																																																				
5	IDE_D7																																																																																				
6	IDE_D8																																																																																				
7	IDE_D6																																																																																				
8	IDE_D9																																																																																				
9	IDE_D5																																																																																				
10	IDE_D10																																																																																				
11	IDE_D4																																																																																				
12	IDE_D11																																																																																				
13	IDE_D3																																																																																				
14	IDE_D12																																																																																				
15	IDE_D2																																																																																				
16	IDE_D13																																																																																				
17	IDE_D1																																																																																				
18	IDE_D14																																																																																				
19	IDE_D0																																																																																				
20	IDE_D15																																																																																				
21	GND																																																																																				
22	N.C.																																																																																				
23	GND																																																																																				
24	IDE_nIOW																																																																																				
25	IDE_nIOR																																																																																				
26	GND																																																																																				
27	IDE_nIORDY																																																																																				
28	GND																																																																																				
29	N.C.																																																																																				
30	IDE_IRQ																																																																																				
31	IDE_ADDR1																																																																																				
32	TPOINT_1																																																																																				
33	IDE_ADDR0																																																																																				
34	IDE_ADDR2																																																																																				
35	IDE_NCS0																																																																																				
36	IDE_NCS1																																																																																				
37	DASP																																																																																				
38	VCC_3P3																																																																																				
39	VCC_3P3																																																																																				
40	IDE_DEVADR																																																																																				
6	CON19	DM1AA-SF-PEJ	SD/MMC connector <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>SD_DATA3</td></tr><tr><td>2</td><td>SD_CMD</td></tr><tr><td>3</td><td>GND</td></tr><tr><td>4</td><td>VCC_3P3</td></tr><tr><td>5</td><td>SD_CLK</td></tr><tr><td>6</td><td>GND</td></tr><tr><td>7</td><td>SD_DATA0</td></tr><tr><td>8</td><td>SD_DATA1</td></tr><tr><td>9</td><td>SD_DATA2</td></tr></table>	Pin Description		1	SD_DATA3	2	SD_CMD	3	GND	4	VCC_3P3	5	SD_CLK	6	GND	7	SD_DATA0	8	SD_DATA1	9	SD_DATA2																																																														
Pin Description																																																																																					
1	SD_DATA3																																																																																				
2	SD_CMD																																																																																				
3	GND																																																																																				
4	VCC_3P3																																																																																				
5	SD_CLK																																																																																				
6	GND																																																																																				
7	SD_DATA0																																																																																				
8	SD_DATA1																																																																																				
9	SD_DATA2																																																																																				

			<table><tr><td>9</td><td>SD_DATA2</td></tr><tr><td>10</td><td>GND</td></tr><tr><td>11</td><td>SD_CD</td></tr><tr><td>12</td><td>SD_WP</td></tr><tr><td>13</td><td>GND</td></tr><tr><td>14</td><td>GND</td></tr></table>	9	SD_DATA2	10	GND	11	SD_CD	12	SD_WP	13	GND	14	GND																																																																																										
9	SD_DATA2																																																																																																								
10	GND																																																																																																								
11	SD_CD																																																																																																								
12	SD_WP																																																																																																								
13	GND																																																																																																								
14	GND																																																																																																								
7	CON20	MI21A-50PD-SF-EJR	CF connector <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>GND</td></tr><tr><td>2</td><td>PD3</td></tr><tr><td>3</td><td>PD4</td></tr><tr><td>4</td><td>PD5</td></tr><tr><td>5</td><td>PD6</td></tr><tr><td>6</td><td>PD7</td></tr><tr><td>7</td><td>nS1_PCE0</td></tr><tr><td>8</td><td>PA10</td></tr><tr><td>9</td><td>InPOE</td></tr><tr><td>10</td><td>PA9</td></tr><tr><td>11</td><td>PA8</td></tr><tr><td>12</td><td>PA7</td></tr><tr><td>13</td><td>SO_VCC</td></tr><tr><td>14</td><td>PA6</td></tr><tr><td>15</td><td>PA5</td></tr><tr><td>16</td><td>PA4</td></tr><tr><td>17</td><td>PA3</td></tr><tr><td>18</td><td>PA2</td></tr><tr><td>19</td><td>PA1</td></tr><tr><td>20</td><td>PA0</td></tr><tr><td>21</td><td>PD0</td></tr><tr><td>22</td><td>PD1</td></tr><tr><td>23</td><td>PD2</td></tr><tr><td>24</td><td>nS1_IOIS16</td></tr><tr><td>25</td><td>nS1CD1</td></tr><tr><td>26</td><td>nS1CD0</td></tr><tr><td>27</td><td>PD11</td></tr><tr><td>28</td><td>PD12</td></tr><tr><td>29</td><td>PD13</td></tr><tr><td>30</td><td>PD14</td></tr><tr><td>31</td><td>PD15</td></tr><tr><td>32</td><td>nS1_PCE1</td></tr><tr><td>33</td><td>S1_VS1</td></tr><tr><td>34</td><td>InPIOR</td></tr><tr><td>35</td><td>InPIOW</td></tr><tr><td>36</td><td>InPWE</td></tr><tr><td>37</td><td>iRDYnBSY</td></tr><tr><td>38</td><td>SO_VCC</td></tr><tr><td>39</td><td>N.C.</td></tr><tr><td>40</td><td>S1_VS2</td></tr><tr><td>41</td><td>nCF1_RESET</td></tr><tr><td>42</td><td>CF_nWAIT</td></tr><tr><td>43</td><td>N.C.</td></tr><tr><td>44</td><td>InPREG</td></tr><tr><td>45</td><td>iS1SPKR</td></tr><tr><td>46</td><td>iS1STSCHG</td></tr><tr><td>47</td><td>PD8</td></tr><tr><td>48</td><td>PD9</td></tr><tr><td>49</td><td>PD10</td></tr><tr><td>50</td><td>GND</td></tr></table>	Pin Description		1	GND	2	PD3	3	PD4	4	PD5	5	PD6	6	PD7	7	nS1_PCE0	8	PA10	9	InPOE	10	PA9	11	PA8	12	PA7	13	SO_VCC	14	PA6	15	PA5	16	PA4	17	PA3	18	PA2	19	PA1	20	PA0	21	PD0	22	PD1	23	PD2	24	nS1_IOIS16	25	nS1CD1	26	nS1CD0	27	PD11	28	PD12	29	PD13	30	PD14	31	PD15	32	nS1_PCE1	33	S1_VS1	34	InPIOR	35	InPIOW	36	InPWE	37	iRDYnBSY	38	SO_VCC	39	N.C.	40	S1_VS2	41	nCF1_RESET	42	CF_nWAIT	43	N.C.	44	InPREG	45	iS1SPKR	46	iS1STSCHG	47	PD8	48	PD9	49	PD10	50	GND
Pin Description																																																																																																									
1	GND																																																																																																								
2	PD3																																																																																																								
3	PD4																																																																																																								
4	PD5																																																																																																								
5	PD6																																																																																																								
6	PD7																																																																																																								
7	nS1_PCE0																																																																																																								
8	PA10																																																																																																								
9	InPOE																																																																																																								
10	PA9																																																																																																								
11	PA8																																																																																																								
12	PA7																																																																																																								
13	SO_VCC																																																																																																								
14	PA6																																																																																																								
15	PA5																																																																																																								
16	PA4																																																																																																								
17	PA3																																																																																																								
18	PA2																																																																																																								
19	PA1																																																																																																								
20	PA0																																																																																																								
21	PD0																																																																																																								
22	PD1																																																																																																								
23	PD2																																																																																																								
24	nS1_IOIS16																																																																																																								
25	nS1CD1																																																																																																								
26	nS1CD0																																																																																																								
27	PD11																																																																																																								
28	PD12																																																																																																								
29	PD13																																																																																																								
30	PD14																																																																																																								
31	PD15																																																																																																								
32	nS1_PCE1																																																																																																								
33	S1_VS1																																																																																																								
34	InPIOR																																																																																																								
35	InPIOW																																																																																																								
36	InPWE																																																																																																								
37	iRDYnBSY																																																																																																								
38	SO_VCC																																																																																																								
39	N.C.																																																																																																								
40	S1_VS2																																																																																																								
41	nCF1_RESET																																																																																																								
42	CF_nWAIT																																																																																																								
43	N.C.																																																																																																								
44	InPREG																																																																																																								
45	iS1SPKR																																																																																																								
46	iS1STSCHG																																																																																																								
47	PD8																																																																																																								
48	PD9																																																																																																								
49	PD10																																																																																																								
50	GND																																																																																																								
8	CON16	Normal USB type	USB1.1 Host connector (type A)																																																																																																						

		A	<table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>VCC_USBH</td></tr><tr><td>2</td><td>USB1.1_D0_HM</td></tr><tr><td>3</td><td>USB1.1_D0_HP</td></tr><tr><td>4</td><td>GND</td></tr></table>	Pin Description		1	VCC_USBH	2	USB1.1_D0_HM	3	USB1.1_D0_HP	4	GND						
Pin Description																			
1	VCC_USBH																		
2	USB1.1_D0_HM																		
3	USB1.1_D0_HP																		
4	GND																		
9	CON14	Mini USB type B	USB1.1 Device connector (type B mini) <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>VCC_USB5</td></tr><tr><td>2</td><td>USB1.1_D1_DM</td></tr><tr><td>3</td><td>USB1.1_D1_DP</td></tr><tr><td>4</td><td>GND</td></tr><tr><td>5</td><td>GND</td></tr></table>	Pin Description		1	VCC_USB5	2	USB1.1_D1_DM	3	USB1.1_D1_DP	4	GND	5	GND				
Pin Description																			
1	VCC_USB5																		
2	USB1.1_D1_DM																		
3	USB1.1_D1_DP																		
4	GND																		
5	GND																		
10	CON13	Mini USB type B	USB2.0 Device connector (type B mini) <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>Connected to PGB0603</td></tr><tr><td>2</td><td>Connected to DM Pin of NET2272</td></tr><tr><td>3</td><td>Connected to DP Pin of NET2272</td></tr><tr><td>4</td><td>GND</td></tr><tr><td>5</td><td>GND</td></tr></table>	Pin Description		1	Connected to PGB0603	2	Connected to DM Pin of NET2272	3	Connected to DP Pin of NET2272	4	GND	5	GND				
Pin Description																			
1	Connected to PGB0603																		
2	Connected to DM Pin of NET2272																		
3	Connected to DP Pin of NET2272																		
4	GND																		
5	GND																		
11	CON6	PJK-631	Audio-Out Earphone Jack <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>N.C.</td></tr><tr><td>2</td><td>GND</td></tr><tr><td>3</td><td>ROUT</td></tr><tr><td>4</td><td>LOUT</td></tr></table>	Pin Description		1	N.C.	2	GND	3	ROUT	4	LOUT						
Pin Description																			
1	N.C.																		
2	GND																		
3	ROUT																		
4	LOUT																		
12	CON8	PJK-631	Audio-In Earphone Jack <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>N.C.</td></tr><tr><td>2</td><td>GND</td></tr><tr><td>3</td><td>RIN</td></tr><tr><td>4</td><td>LIN</td></tr></table>	Pin Description		1	N.C.	2	GND	3	RIN	4	LIN						
Pin Description																			
1	N.C.																		
2	GND																		
3	RIN																		
4	LIN																		
13	CON12	PJK-631	Video-In Jack <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>CVBS_IN</td></tr><tr><td>2</td><td>GND</td></tr><tr><td>3</td><td>C_IN</td></tr><tr><td>4</td><td>Y_IN</td></tr></table>	Pin Description		1	CVBS_IN	2	GND	3	C_IN	4	Y_IN						
Pin Description																			
1	CVBS_IN																		
2	GND																		
3	C_IN																		
4	Y_IN																		
14	SW1	PB_SW STM-1101H	Power button for momentary-on/off function																
15	SW4	HSS165A	5-way navigation switch <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>DIRECTION_1_S</td></tr><tr><td>2</td><td>DIRECTION_2_S</td></tr><tr><td>3</td><td>DIRECTION_3_S</td></tr><tr><td>4</td><td>DIRECTION_4_S</td></tr><tr><td>5</td><td>GND</td></tr><tr><td>6</td><td>CENTER_S</td></tr><tr><td>7</td><td>GND</td></tr></table>	Pin Description		1	DIRECTION_1_S	2	DIRECTION_2_S	3	DIRECTION_3_S	4	DIRECTION_4_S	5	GND	6	CENTER_S	7	GND
Pin Description																			
1	DIRECTION_1_S																		
2	DIRECTION_2_S																		
3	DIRECTION_3_S																		
4	DIRECTION_4_S																		
5	GND																		
6	CENTER_S																		
7	GND																		
16	SW7	SSSS7A0101	Slide switch for function Lock/Release																
17	SW3	PB_SW STM-1101H	Push button for user's purpose																
18	SW5	PB_SW STM-	Push button for user's purpose																

		1101H																																																																	
19	SW6	PB_SW STM-1101H	Push button for user's purpose																																																																
20	CON2	53014-0710	JTAG connector <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>VCC_3P3</td></tr><tr><td>2</td><td>JTAG_NTRST</td></tr><tr><td>3</td><td>JTAG_TDI</td></tr><tr><td>4</td><td>JTAG_TMS</td></tr><tr><td>5</td><td>JTAG_TCK</td></tr><tr><td>6</td><td>JTAG_TDO</td></tr></table>	Pin Description		1	VCC_3P3	2	JTAG_NTRST	3	JTAG_TDI	4	JTAG_TMS	5	JTAG_TCK	6	JTAG_TDO																																																		
Pin Description																																																																			
1	VCC_3P3																																																																		
2	JTAG_NTRST																																																																		
3	JTAG_TDI																																																																		
4	JTAG_TMS																																																																		
5	JTAG_TCK																																																																		
6	JTAG_TDO																																																																		
21	CON18	53014-10	UART port for debugging (VR3511F UART channel 1) <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>N.C.</td></tr><tr><td>2</td><td>Connected to R1IN Pin of MAX3232</td></tr><tr><td>3</td><td>Connected to T1OUT Pin of MAX3232</td></tr><tr><td>4</td><td>GND</td></tr></table>	Pin Description		1	N.C.	2	Connected to R1IN Pin of MAX3232	3	Connected to T1OUT Pin of MAX3232	4	GND																																																						
Pin Description																																																																			
1	N.C.																																																																		
2	Connected to R1IN Pin of MAX3232																																																																		
3	Connected to T1OUT Pin of MAX3232																																																																		
4	GND																																																																		
22	CON15	12512WS-04	UART port for GPS (VR3511F UART channel 0) <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>GND</td></tr><tr><td>2</td><td>GPS_DI</td></tr><tr><td>3</td><td>GPS_DO</td></tr><tr><td>4</td><td>VCC_LCD5</td></tr></table>	Pin Description		1	GND	2	GPS_DI	3	GPS_DO	4	VCC_LCD5																																																						
Pin Description																																																																			
1	GND																																																																		
2	GPS_DI																																																																		
3	GPS_DO																																																																		
4	VCC_LCD5																																																																		
23	CON1	HEADER 22x2	External board connector <table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>WDATA0</td></tr><tr><td>2</td><td>WDATA1</td></tr><tr><td>3</td><td>WDATA2</td></tr><tr><td>4</td><td>WDATA3</td></tr><tr><td>5</td><td>WDATA4</td></tr><tr><td>6</td><td>WDATA5</td></tr><tr><td>7</td><td>WDATA6</td></tr><tr><td>8</td><td>WDATA7</td></tr><tr><td>9</td><td>WDATA8</td></tr><tr><td>10</td><td>WDATA9</td></tr><tr><td>11</td><td>WDATA10</td></tr><tr><td>12</td><td>WDATA11</td></tr><tr><td>13</td><td>WDATA12</td></tr><tr><td>14</td><td>WDATA13</td></tr><tr><td>15</td><td>WDATA14</td></tr><tr><td>16</td><td>WDATA15</td></tr><tr><td>17</td><td>ETHER_nCS</td></tr><tr><td>18</td><td>WnWE</td></tr><tr><td>19</td><td>RESET</td></tr><tr><td>20</td><td>WnOE</td></tr><tr><td>21</td><td>ETHERNET_IRQ</td></tr><tr><td>22</td><td>VCC_3P3</td></tr><tr><td>23</td><td>ETHER_IORDY</td></tr><tr><td>24</td><td>EXTUART_nCSA</td></tr><tr><td>25</td><td>WnBTENB1</td></tr><tr><td>26</td><td>EXTUART_nCSB</td></tr><tr><td>27</td><td>VCC_3P3</td></tr><tr><td>28</td><td>EXTUART_nCSC</td></tr><tr><td>29</td><td>N.C.</td></tr><tr><td>30</td><td>EXTUART_nCSD</td></tr><tr><td>31</td><td>N.C.</td></tr></table>	Pin Description		1	WDATA0	2	WDATA1	3	WDATA2	4	WDATA3	5	WDATA4	6	WDATA5	7	WDATA6	8	WDATA7	9	WDATA8	10	WDATA9	11	WDATA10	12	WDATA11	13	WDATA12	14	WDATA13	15	WDATA14	16	WDATA15	17	ETHER_nCS	18	WnWE	19	RESET	20	WnOE	21	ETHERNET_IRQ	22	VCC_3P3	23	ETHER_IORDY	24	EXTUART_nCSA	25	WnBTENB1	26	EXTUART_nCSB	27	VCC_3P3	28	EXTUART_nCSC	29	N.C.	30	EXTUART_nCSD	31	N.C.
Pin Description																																																																			
1	WDATA0																																																																		
2	WDATA1																																																																		
3	WDATA2																																																																		
4	WDATA3																																																																		
5	WDATA4																																																																		
6	WDATA5																																																																		
7	WDATA6																																																																		
8	WDATA7																																																																		
9	WDATA8																																																																		
10	WDATA9																																																																		
11	WDATA10																																																																		
12	WDATA11																																																																		
13	WDATA12																																																																		
14	WDATA13																																																																		
15	WDATA14																																																																		
16	WDATA15																																																																		
17	ETHER_nCS																																																																		
18	WnWE																																																																		
19	RESET																																																																		
20	WnOE																																																																		
21	ETHERNET_IRQ																																																																		
22	VCC_3P3																																																																		
23	ETHER_IORDY																																																																		
24	EXTUART_nCSA																																																																		
25	WnBTENB1																																																																		
26	EXTUART_nCSB																																																																		
27	VCC_3P3																																																																		
28	EXTUART_nCSC																																																																		
29	N.C.																																																																		
30	EXTUART_nCSD																																																																		
31	N.C.																																																																		

				31	N.C.	
				32	N.C.	
				33	WADDR1	
				34	WADDR2	
				35	WADDR3	
				36	N.C.	
				37	N.C.	
				38	N.C.	
				39	QUART_INTA	
				40	QUART_INTB	
				41	QUART_INTC	
				42	QUART_INTD	
				43	GND	
				44	GND	

<Table 20. Connector & Switch description of Main Board>

4.2.2. External Board

- IC components

No	Ref No	Part Name	Description
1	U30	ST16C554CQ64	Quad UART IC

<Table 21. IC Components description of External Board>

- Connector & Switch components

No	Ref No	Part Name	Description																																																										
1	CON22	HEADER 22x2	External board connector																																																										
<table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>XZD0</td></tr><tr><td>2</td><td>XZD1</td></tr><tr><td>3</td><td>XZD2</td></tr><tr><td>4</td><td>XZD3</td></tr><tr><td>5</td><td>XZD4</td></tr><tr><td>6</td><td>XZD5</td></tr><tr><td>7</td><td>XZD6</td></tr><tr><td>8</td><td>XZD7</td></tr><tr><td>9</td><td>XZD8</td></tr><tr><td>10</td><td>XZD9</td></tr><tr><td>11</td><td>XZD10</td></tr><tr><td>12</td><td>XZD11</td></tr><tr><td>13</td><td>XZD12</td></tr><tr><td>14</td><td>XZD13</td></tr><tr><td>15</td><td>XZD14</td></tr><tr><td>16</td><td>XZD15</td></tr><tr><td>17</td><td>XETHER_nCS</td></tr><tr><td>18</td><td>XnWE</td></tr><tr><td>19</td><td>XRST</td></tr><tr><td>20</td><td>XnOE</td></tr><tr><td>21</td><td>XETHER_IRQ</td></tr><tr><td>22</td><td>VCC_EX3P3</td></tr><tr><td>23</td><td>XETHER_RDY</td></tr><tr><td>24</td><td>XnCSA</td></tr><tr><td>25</td><td>XnBTENB1</td></tr><tr><td>26</td><td>XnCSB</td></tr><tr><td>27</td><td>VCC_EX3P3</td></tr><tr><td>28</td><td>XnCSC</td></tr></table>				Pin Description		1	XZD0	2	XZD1	3	XZD2	4	XZD3	5	XZD4	6	XZD5	7	XZD6	8	XZD7	9	XZD8	10	XZD9	11	XZD10	12	XZD11	13	XZD12	14	XZD13	15	XZD14	16	XZD15	17	XETHER_nCS	18	XnWE	19	XRST	20	XnOE	21	XETHER_IRQ	22	VCC_EX3P3	23	XETHER_RDY	24	XnCSA	25	XnBTENB1	26	XnCSB	27	VCC_EX3P3	28	XnCSC
Pin Description																																																													
1	XZD0																																																												
2	XZD1																																																												
3	XZD2																																																												
4	XZD3																																																												
5	XZD4																																																												
6	XZD5																																																												
7	XZD6																																																												
8	XZD7																																																												
9	XZD8																																																												
10	XZD9																																																												
11	XZD10																																																												
12	XZD11																																																												
13	XZD12																																																												
14	XZD13																																																												
15	XZD14																																																												
16	XZD15																																																												
17	XETHER_nCS																																																												
18	XnWE																																																												
19	XRST																																																												
20	XnOE																																																												
21	XETHER_IRQ																																																												
22	VCC_EX3P3																																																												
23	XETHER_RDY																																																												
24	XnCSA																																																												
25	XnBTENB1																																																												
26	XnCSB																																																												
27	VCC_EX3P3																																																												
28	XnCSC																																																												

				<table><tr><td>28</td><td>XnCSC</td></tr><tr><td>29</td><td>N.C.</td></tr><tr><td>30</td><td>XnCSD</td></tr><tr><td>31</td><td>N.C.</td></tr><tr><td>32</td><td>N.C.</td></tr><tr><td>33</td><td>XZA1</td></tr><tr><td>34</td><td>XZA2</td></tr><tr><td>35</td><td>XZA3</td></tr><tr><td>36</td><td>N.C.</td></tr><tr><td>37</td><td>N.C.</td></tr><tr><td>38</td><td>N.C.</td></tr><tr><td>39</td><td>XINTA</td></tr><tr><td>40</td><td>XINTB</td></tr><tr><td>41</td><td>XINTC</td></tr><tr><td>42</td><td>XINTD</td></tr><tr><td>43</td><td>GND</td></tr><tr><td>44</td><td>GND</td></tr></table>	28	XnCSC	29	N.C.	30	XnCSD	31	N.C.	32	N.C.	33	XZA1	34	XZA2	35	XZA3	36	N.C.	37	N.C.	38	N.C.	39	XINTA	40	XINTB	41	XINTC	42	XINTD	43	GND	44	GND																													
28	XnCSC																																																																		
29	N.C.																																																																		
30	XnCSD																																																																		
31	N.C.																																																																		
32	N.C.																																																																		
33	XZA1																																																																		
34	XZA2																																																																		
35	XZA3																																																																		
36	N.C.																																																																		
37	N.C.																																																																		
38	N.C.																																																																		
39	XINTA																																																																		
40	XINTB																																																																		
41	XINTC																																																																		
42	XINTD																																																																		
43	GND																																																																		
44	GND																																																																		
2	CON23	HEADER 15x2	Ethernet board connector	<table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>XZD0</td></tr><tr><td>2</td><td>XZD1</td></tr><tr><td>3</td><td>XZD2</td></tr><tr><td>4</td><td>XZD3</td></tr><tr><td>5</td><td>XZD4</td></tr><tr><td>6</td><td>XZD5</td></tr><tr><td>7</td><td>XZD6</td></tr><tr><td>8</td><td>XZD7</td></tr><tr><td>9</td><td>XZD8</td></tr><tr><td>10</td><td>XZD9</td></tr><tr><td>11</td><td>XZD10</td></tr><tr><td>12</td><td>XZD11</td></tr><tr><td>13</td><td>XZD12</td></tr><tr><td>14</td><td>XnWE</td></tr><tr><td>15</td><td>XETHER_nCS</td></tr><tr><td>16</td><td>XnOE</td></tr><tr><td>17</td><td>XETHER_RST</td></tr><tr><td>18</td><td>XETHER_RDY</td></tr><tr><td>19</td><td>XETHER_IRQ</td></tr><tr><td>20</td><td>XZA1</td></tr><tr><td>21</td><td>XZA2</td></tr><tr><td>22</td><td>XZA3</td></tr><tr><td>23</td><td>XZD13</td></tr><tr><td>24</td><td>XZD14</td></tr><tr><td>25</td><td>XZD15</td></tr><tr><td>26</td><td>XnBTENB1</td></tr><tr><td>27</td><td>GND</td></tr><tr><td>28</td><td>VCC_EX3P3</td></tr><tr><td>29</td><td>GND</td></tr><tr><td>30</td><td>VCC_EX3P3</td></tr></table>	Pin Description		1	XZD0	2	XZD1	3	XZD2	4	XZD3	5	XZD4	6	XZD5	7	XZD6	8	XZD7	9	XZD8	10	XZD9	11	XZD10	12	XZD11	13	XZD12	14	XnWE	15	XETHER_nCS	16	XnOE	17	XETHER_RST	18	XETHER_RDY	19	XETHER_IRQ	20	XZA1	21	XZA2	22	XZA3	23	XZD13	24	XZD14	25	XZD15	26	XnBTENB1	27	GND	28	VCC_EX3P3	29	GND	30	VCC_EX3P3	
Pin Description																																																																			
1	XZD0																																																																		
2	XZD1																																																																		
3	XZD2																																																																		
4	XZD3																																																																		
5	XZD4																																																																		
6	XZD5																																																																		
7	XZD6																																																																		
8	XZD7																																																																		
9	XZD8																																																																		
10	XZD9																																																																		
11	XZD10																																																																		
12	XZD11																																																																		
13	XZD12																																																																		
14	XnWE																																																																		
15	XETHER_nCS																																																																		
16	XnOE																																																																		
17	XETHER_RST																																																																		
18	XETHER_RDY																																																																		
19	XETHER_IRQ																																																																		
20	XZA1																																																																		
21	XZA2																																																																		
22	XZA3																																																																		
23	XZD13																																																																		
24	XZD14																																																																		
25	XZD15																																																																		
26	XnBTENB1																																																																		
27	GND																																																																		
28	VCC_EX3P3																																																																		
29	GND																																																																		
30	VCC_EX3P3																																																																		
3	CON24	53014-10	External UART A connector (10 pin)	<table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>VCC_EX3P3</td></tr><tr><td>2</td><td>nCDA</td></tr><tr><td>3</td><td>RXA</td></tr><tr><td>4</td><td>TXA</td></tr><tr><td>5</td><td>nDTRA</td></tr><tr><td>6</td><td>EXGND</td></tr><tr><td>7</td><td>nDSRA</td></tr><tr><td>8</td><td>nRTSA</td></tr></table>	Pin Description		1	VCC_EX3P3	2	nCDA	3	RXA	4	TXA	5	nDTRA	6	EXGND	7	nDSRA	8	nRTSA																																													
Pin Description																																																																			
1	VCC_EX3P3																																																																		
2	nCDA																																																																		
3	RXA																																																																		
4	TXA																																																																		
5	nDTRA																																																																		
6	EXGND																																																																		
7	nDSRA																																																																		
8	nRTSA																																																																		

				8	nRTSA											
				9	nCTSA											
				10	nRIA											
4	CON25	53014-04	External UART B connector (4 pin)													
			<table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>VCC_EX3P3</td></tr><tr><td>2</td><td>RXB</td></tr><tr><td>3</td><td>TXB</td></tr><tr><td>4</td><td>EXGND</td></tr></table>				Pin Description		1	VCC_EX3P3	2	RXB	3	TXB	4	EXGND
Pin Description																
1	VCC_EX3P3															
2	RXB															
3	TXB															
4	EXGND															
5	CON26	53014-04	External UART B connector (4 pin)													
			<table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>VCC_EX3P3</td></tr><tr><td>2</td><td>RXC</td></tr><tr><td>3</td><td>TXC</td></tr><tr><td>4</td><td>EXGND</td></tr></table>				Pin Description		1	VCC_EX3P3	2	RXC	3	TXC	4	EXGND
Pin Description																
1	VCC_EX3P3															
2	RXC															
3	TXC															
4	EXGND															
6	CON27	53014-04	External UART B connector (4 pin)													
			<table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>VCC_EX3P3</td></tr><tr><td>2</td><td>RXD</td></tr><tr><td>3</td><td>TXD</td></tr><tr><td>4</td><td>EXGND</td></tr></table>				Pin Description		1	VCC_EX3P3	2	RXD	3	TXD	4	EXGND
Pin Description																
1	VCC_EX3P3															
2	RXD															
3	TXD															
4	EXGND															

<Table 22. Connector & Switch description of External Board>

4.2.3. Ethernet Board

- IC components

No	Ref No	Part Name	Description
1	-	CS8900A-CQ	Ethernet Controller
2	-	-	Ethernet Transformer

<Table 23. IC Components description of Ethernet Board>

- Connector & Switch components

No	Ref No	Part Name	Description																																		
1	-	HEADER 15x2	Ethernet connector																																		
			<table><tr><th colspan="2">Pin Description</th></tr><tr><td>1</td><td>XZD0</td></tr><tr><td>2</td><td>XZD1</td></tr><tr><td>3</td><td>XZD2</td></tr><tr><td>4</td><td>XZD3</td></tr><tr><td>5</td><td>XZD4</td></tr><tr><td>6</td><td>XZD5</td></tr><tr><td>7</td><td>XZD6</td></tr><tr><td>8</td><td>XZD7</td></tr><tr><td>9</td><td>XZD8</td></tr><tr><td>10</td><td>XZD9</td></tr><tr><td>11</td><td>XZD10</td></tr><tr><td>12</td><td>XZD11</td></tr><tr><td>13</td><td>XZD12</td></tr><tr><td>14</td><td>XnWE</td></tr><tr><td>15</td><td>XETHER_nCS</td></tr><tr><td>16</td><td>XnOE</td></tr></table>	Pin Description		1	XZD0	2	XZD1	3	XZD2	4	XZD3	5	XZD4	6	XZD5	7	XZD6	8	XZD7	9	XZD8	10	XZD9	11	XZD10	12	XZD11	13	XZD12	14	XnWE	15	XETHER_nCS	16	XnOE
Pin Description																																					
1	XZD0																																				
2	XZD1																																				
3	XZD2																																				
4	XZD3																																				
5	XZD4																																				
6	XZD5																																				
7	XZD6																																				
8	XZD7																																				
9	XZD8																																				
10	XZD9																																				
11	XZD10																																				
12	XZD11																																				
13	XZD12																																				
14	XnWE																																				
15	XETHER_nCS																																				
16	XnOE																																				

				16	XnOE	
				17	XETHER_RST	
				18	XETHER_RDY	
				19	XETHER_IRQ	
				20	XZA1	
				21	XZA2	
				22	XZA3	
				23	XZD13	
				24	XZD14	
				25	XZD15	
				26	XnBTENB1	
				27	GND	
				28	VCC_EX3P3	
				29	GND	
				30	VCC_EX3P3	
2	-	-	Ethernet cable connector			

<Table 24. Connector & Switch description of Ethernet Board>